MitraClip™ G5 SYSTEM

Rx ONLY

MitraClip™ G5 Delivery System
CDS0801-NT
CDS0801-NTW
CDS0801-XT
CDS0801-XTW

MitraClip™ G5 Steerable Guide Catheter
SGC0801

MitraClip™ and TriClip™ ACCESSORIES
Stabilizer SZR08
Lift LFT07 / LFT01ST
Support Plate PLT07 / PLT01ST

INSTRUCTIONS FOR USE

WARNING: Use of the MitraClip™ G5 System should be restricted to those physicians trained to perform invasive endovascular and transseptal procedures and to those physicians trained in the proper use of the system.

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1.0 DEVICE DESCRIPTION

The MitraClip™ G5 System is comprised of the MitraClip™ G5 Steerable Guide Catheter (SGC) and the Clip Delivery System (CDS) and the MitraClip™ and TriClip™ Accessories.

MitraClip™ G5 Steerable Guide Catheter

The MitraClip™ G5 Steerable Guide Catheter's primary function is to dilate the atrial septum, maneuver to the target location above the mitral valve and position the Clip Delivery System.

MitraClip™ G5 Delivery System

The MitraClip™ G5 Delivery System (CDS) is used to deliver, position, and place the Implant of the MitraClip™ G5 System on the mitral valve leaflets. The CDS is comprised of the Delivery Catheter, the Steerable Sleeve, a handle, and the MitraClip™ G5 Implant. The user interface on the CDS allows for the adjustment of the Implant Clip to any position (open, closed, inverted) for implantation.

The Implant grasps and coapts the mitral valve leaflets resulting in fixed approximation of the leaflets throughout the cardiac cycle. It is available in four sizes (NT, XT, NTW, XTW) and can be locked, unlocked, and repeatedly opened and closed to allow for repositioning of the Implant to the target location.

Implant size selection should be determined by the heart team and optimized based on patient anatomy and condition.

Accessories

The Accessories are intended to support the MitraClip™ G5 System during the procedure. The Accessories consist of the Stabilizer, the Support Plate, and the Lift. The Lift and the Support Plate are used outside of the sterile field to provide a stable platform during the procedure. The Stabilizer is used in the sterile field to support and position the Steerable Guide Catheter and MitraClip™ Delivery System during the procedure.

Figure 1 shows the MitraClip™ G5 Implant Positions. **Table 1** lists the MitraClip™ G5 System Dimensions.

Figure 1: MitraClip™ G5 Implant Positions

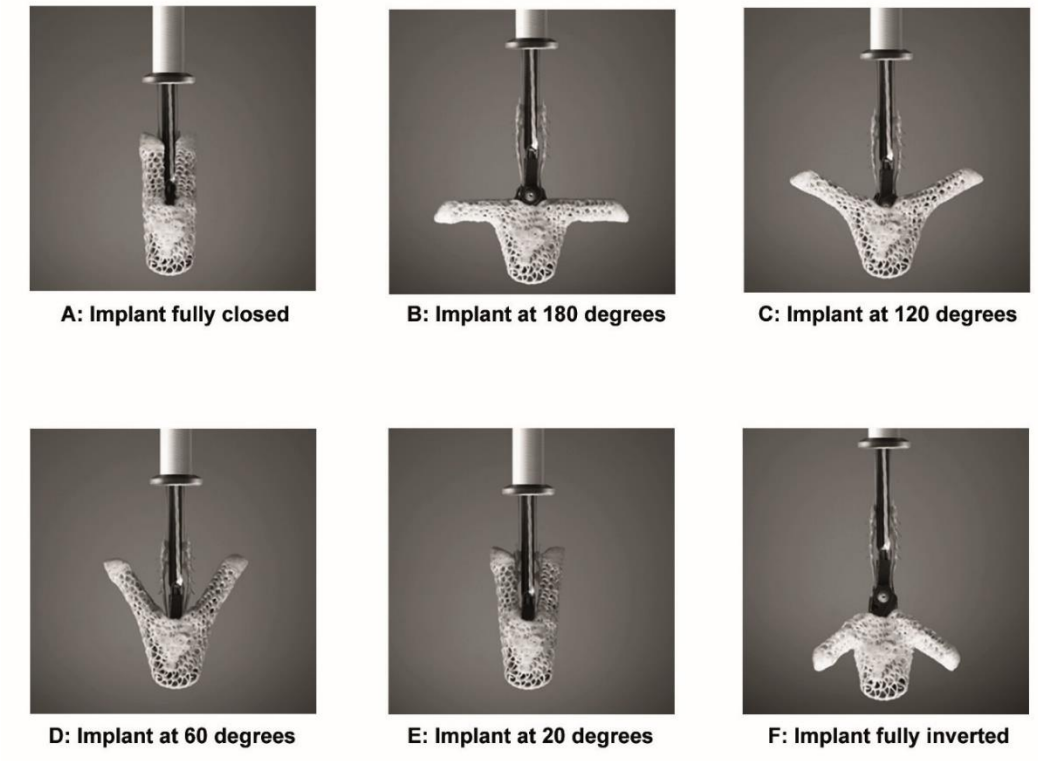


Table 1: MitraClip™ G5 System Dimensions

Component	***Dimension***			
Delivery Catheter				
Extended Length (from Sleeve curved at 90 degrees)	*≥ 65 mm*			
Steerable Sleeve				
Working Length	*109.5 cm*			
Catheter Distal Shaft Outer Diameter	*5.3 mm (16 Fr)*			
MitraClip™ G5 Implant Sizes	*NT*	*NTW*	*XT*	*XTW*
Grasping Width at 120 degrees, *minimum* mm	*17 mm*		*22 mm*	
Implant Width at 180 degrees, *nominal* mm	*20 mm*		*25 mm*	
Arm Width, *maximum* mm	*4 mm*	*6 mm*	*4mm*	*6 mm*
Maximum Arm (Coaptation) Length, *maximum* mm	*9 mm*		*12 mm*	
MitraClip™ G5 Steerable Guide Catheter				
Working Length	*78.0 cm*			
Catheter Proximal Shaft Outer Diameter	*8.4 mm (25 Fr)*			
Catheter Distal Shaft / Tip Outer Diameter (Septal Crossing)	*7.7 mm (23 Fr)*			
Dilator				
Working Length	*122.0 cm*			
Shaft Inner Diameter	*1.0 mm (3 Fr)*			
Shaft Outer Diameter	*5.4 mm (16 Fr)*			
Distal Tip Outer Diameter	*1.5 mm (4 Fr)*			

2.0 INDICATION FOR USE

The MitraClip™ G5 System is indicated for the percutaneous reduction of significant symptomatic mitral regurgitation (MR ≥ 3+) due to primary abnormality of the mitral apparatus [degenerative MR] in patients who have been determined to be at prohibitive risk for mitral valve surgery by a heart team, which includes a cardiac surgeon experienced in mitral valve surgery and a cardiologist experienced in mitral valve disease, and in whom existing comorbidities would not preclude the expected benefit from reduction of the mitral regurgitation.

The MitraClip™ G5 System, when used with maximally tolerated guideline-directed medical therapy (GDMT), is indicated for the treatment of symptomatic, moderate-to-severe or severe secondary (or functional) mitral regurgitation (MR; MR ≥ Grade III per American Society of Echocardiography criteria) in patients with a left ventricular ejection fraction (LVEF) ≥ 20% and ≤ 50%, and a left ventricular end systolic dimension (LVESD) ≤ 70 mm whose symptoms and MR severity persist despite maximally tolerated GDMT as determined by a multidisciplinary heart team experienced in the evaluation and treatment of heart failure and mitral valve disease.

3.0 CONTRAINDICATIONS

The MitraClip™ G5 System is contraindicated for use in patients with the following conditions:

- Intolerance, including allergy or untreatable hypersensitivity, to procedural anticoagulation
- Untreatable hypersensitivity to Implant components (nickel- titanium alloy, cobalt-chromium alloy)
- Active endocarditis or other active infection of the mitral valve

4.0 WARNINGS

- The MitraClip™ G5 Procedure should be considered with caution in patients with rheumatic disease who have significant leaflet thickening and small annular dimensions considering the risk of iatrogenic mitral stenosis**.**
- The MitraClip™ G5 Steerable Guide Catheter and MitraClip™ G5 Delivery System are designed, intended, and distributed for single use only. Cleaning, re-sterilization and / or re-use may result in infections, malfunction of the device and other serious injury or death.
- The Stabilizer, Lift, and Support Plate are non-sterile and must be cleaned, disinfected, and/or sterilized prior to each use. Follow the cleaning, disinfection, and sterilization instructions provided with these accessories.
- The devices should be handled using standard sterile technique to prevent infection.
- Do not use the devices if the ‘Use by’ date specified on the package has elapsed.
- Do not use the devices if the package is damaged or the packing seal is broken.
- Inspect all reusable accessories prior to use. Do not use if the devices are damaged or mishandled.
- Read all instructions carefully. Use universal precautions for biohazards and sharps while handling the MitraClip™ G5 System to avoid user injury.
- Failure to prepare the device as stated in these instructions and failure to handle the device with care might result in damage to the device coating, which may lead to additional intervention or serious adverse events.

- Failure to follow these instructions, warnings and precautions may lead to device damage, user injury, or patient injury including:
  - Failure to deliver MitraClip™ G5 Implant to the intended site
  - Difficulty or failure to retrieve MitraClip™ G5 System components

5.0 PRECAUTIONS

- The MitraClip™ G5 Implant should be implanted with sterile techniques using standard imaging techniques.
- The safety and effectiveness of the MitraClip™ G5 System has not been established in the following patient populations:
  - Pregnant or lactating women
  - Pediatric patients

6.0 PATIENT SELECTION

Patient selection should be performed by a multi-disciplinary heart team. It is advised that the heart team assess whether the use of MitraClip™ G5 System is preferable to surgery or other available treatment options. It is advised that the heart team specifically consider the following when assessing patients for the use of the MitraClip™ G5 System:

For patients with Primary Mitral Regurgitation (PMR), the heart team should include a cardiac surgeon experienced in mitral valve surgery and a cardiologist experienced in mitral valve disease and may also include appropriate physicians to assess valvular anatomy, Clip suitability, and surgical risk:

- Moderate surgical risk is defined to be based on age ≥ 75, STS Risk Repair Score ≥ 2%, or frailty indices or co-morbidities not captured by STS.
- High / Prohibitive surgical risk is defined to be based on STS Risk Repair Score ≥ 6%, STS Risk Replacement Score ≥ 8%, or frailty indices or co-morbidities not captured by STS.

For patients with secondary mitral regurgitation (SMR), the heart team should be experienced in the evaluation and treatment of heart failure and mitral valve disease and determine that symptoms and MR severity persist despite maximally tolerated GDMT.

Patients with the following conditions may have an increased risk of serious adverse events:

- Known or suspected unstable angina or myocardial infarction within the last 12 weeks
- Recent cerebrovascular events
- Rotated heart due to prior cardiac surgery
- Persistent bacteremia despite antibiotic treatment

Use caution when treating patients with hemodynamic instability requiring inotropic support or mechanical heart assistance due to the increased risk of mortality in this patient population. The safety and effectiveness of the MitraClip™ Therapy in these patients has not been evaluated.

7.0 POTENTIAL COMPLICATIONS AND ADVERSE EVENTS

The following events have been identified as possible complications of the MitraClip™ G5 Procedure.

- Allergic reactions or hypersensitivity to latex, contrast agent, anaesthesia, device materials, and drug reactions to anticoagulation, or antiplatelet drugs
- Additional treatment/surgery from device-related complications
- Atrial septal defect
- Bleeding
- Blood disorders (including coagulopathy, hemolysis, and Heparin Induced Thrombocytopenia (HIT))
- Cardiac arrhythmias (including conduction disorders, atrial arrhythmias, ventricular arrhythmias)
- Cardiac ischemic conditions (including myocardial infarction, myocardial ischemia, unstable angina, and stable angina)
- Cardiac perforation
- Cardiac tamponade
- Chest pain
- Death
- Dyspnea
- Edema
- Embolization (device or components of the device)
- Endocarditis
- Fever or hyperthermia
- Fluoroscopy and Transesophageal echocardiogram (TEE) -related complications:
  - Skin injury or tissue changes due to exposure to ionizing radiation
  - Esophageal irritation
  - Esophageal perforation
  - Gastrointestinal bleeding
- Hypotension / hypertension
- Infection including:
  - Septicemia
- Mitral valve complications, which may complicate or prevent later surgical repair, including:
  - Chordal entanglement / rupture
  - Single leaflet device attachment (SLDA)
  - Dislodgement of previously implanted devices
  - Tissue damage
  - Mitral valve stenosis
  - Worsening, persistent or residual regurgitation
- Nausea or vomiting
- Pain
- Pericardial effusion
- Stroke / cerebrovascular accident (CVA) and transient ischemic attack (TIA)
- System organ failure:
  - Cardio-respiratory arrest
  - Worsening heart failure
  - Pulmonary congestion
  - Respiratory dysfunction or failure or atelectasis
  - Renal insufficiency or failure
  - Shock (including cardiogenic and anaphylactic)
- Thrombosis
- Vascular access complications which may require additional intervention, including:
  - Wound dehiscence
  - Bleeding of the access site
  - Arteriovenous fistula, pseudoaneurysm, aneurysm, dissection, perforation (rupture), vascular occlusion
  - Embolism (air, thrombus)
  - Peripheral nerve injury
- Venous thrombosis (including deep vein thrombosis) and thromboembolism (including pulmonary embolism)

8.0 MRI SAFETY INFORMATION



A patient with the MitraClip™ G5 Implant may be safely scanned under the following conditions. Failure to follow these conditions may result in injury to the patient or device malfunction.

Name/Identification of the Device	MitraClip™ G5 Implant
Nominal Value(s) of Static Magnetic Field [T]	1.5T or 3.0T
Maximum Spatial Field Gradient [T/m and gauss/cm]	40 T/m (4,000 gauss/cm)
RF Excitation	Circularly Polarized (CP)
Operating Mode	Normal Operating Mode
Maximum Whole Body SAR [W/kg]	2W/kg
Limits on Scan Duration	2 W/kg whole body average SAR for 60 minutes of continuous RF (a sequence or back-to-back series/scan without breaks) (Under these scan conditions, the implants are expected to produce a maximum temperature rise of less than or equal to 2.7°C after 15 minutes of continuous scanning.)
MR Image Artifact	In non-clinical testing, the image artifact caused by a pair of implants extends approximately 40 mm beyond the implants when imaged with a spin echo or gradient echo pulse sequence in a 3 T magnetic resonance imaging system. The presence of additional implants in a patient's valve may increase the image artifact size when imaged in an MRI system.
If information about a specific parameter is not included, there are no conditions associated with the parameter.	

9.0 INFORMATION TO BE SUPPLIED TO THE PATIENT

Physicians should consider the following information when counseling patients about the MitraClip™ G5 Implant and procedure:

- The potential benefits and risks of the MitraClip™ G5 System.
- Alternative therapies available to the patients and their advantages and disadvantages.

The following are measures that the patient may have to take following the MitraClip™ G5 Procedure:

- Patients undergoing any procedures known to potentially be associated with bacteremia after implantation of the MitraClip™ G5 Implant should be prescribed prophylactic antibiotic therapy prior to such procedures.
- Anticoagulation and other medical therapy should be prescribed per institutional guidelines.
- Patients should be advised to limit strenuous physical activity for at least the first month post-procedure or longer if warranted.

After placement of a MitraClip™ G5 Implant, the Patient Implant Card should be filled out, and the patient should be instructed to carry it at all times.

Patient Guides are available to each site and should be given to the patient to inform them of the risks and benefits of the procedure and alternative treatments. Additional counseling information can be found in the Patient Guide and in the Clinical Data section.

10.0 SPECIAL FACILITIES

Facilities implanting the MitraClip™ G5 Implant need to maintain a plan for treating the patient if cardiovascular surgical intervention is required.

11.0 HOW SUPPLIED

11.1 Contents

One (1) MitraClip™ G5 Delivery System with the MitraClip™ G5 Implant, one (1) Implant Patient Card

11.2 Sterile

The MitraClip™ G5 Delivery System and MitraClip™ G5 Steerable Guide Catheter are individually packaged and sterilized with ethylene oxide gas and provided in a tray with lid, in a sealed pouch. Parts of the devices that are in either direct or indirect contact with circulating blood are non-pyrogenic.

This single use device cannot be reused on another patient, as it is not designed to perform as intended after the first usage. Changes in mechanical, physical, and / or chemical characteristics introduced under conditions of repeated use, cleaning, and / or resterilization may compromise the integrity of the design and / or materials, leading to contamination due to narrow gaps and / or spaces and diminished safety and / or performance of the device. Absence of original labeling may lead to misuse and eliminate traceability. Absence of original packaging may lead to device damage, loss of sterility, and risk of injury to the patient and / or user. Inspect all product prior to use. Do not use if the package is open or damaged, or if product is damaged. Do not reinsert the CDS or Implant after single use in the patient.

11.3 Non-Sterile

WARNING: The Stabilizer, Support Plate and Lift are provided non-sterile. Follow the cleaning and sterilization instructions provided with the Stabilizer, Support Plate and Lift.

12.0 STORAGE

Handle with care. Store in original packaging. Keep dry. Keep away from sunlight.

13.0 REQUIRED ACCESSORIES

SZR08: One (1) Stabilizer
LFT01ST or LFT07: One (1) Lift
PLT01ST or PLT07: One (1) Support Plate
One (1) Silicone Pad (included sterile with the Steerable Guide Catheter)

14.0 ADDITIONAL REQUIRED EQUIPMENT NOT INCLUDED

Transseptal sheath and guidewire
Transseptal needle
Exchange length supportive guidewire
High-pressure three-way stopcocks (4)
Arterial high pressure extension tubing (2)
50-60 cc syringes with luer fitting (2)
1000 ml pressure bag (1)
Sterile IV tubing with thumbwheel occluders (1)
Heparinized sterile saline solution (1) 1 liter bag
Rolling IV Pole
Sterile Basin

15.0 PATIENT PREPARATION

15.1 Prepare the patient per institution's standard practice for transseptal catheterization.

15.2 Place support plate under patient's leg in the region between the area of the upper leg and the knee and place the Lift over the ipsilateral lower extremity prior to draping the patient.

15.3 Place the Lift on the Support Plate such that the front edge (i.e., the edge that corresponds with the shorter legs of the Lift) is approximately 80 cm from the patient's mid sternum.

15.4 Adjust the height of the Lift so that the front edge of the Lift is close to the patient's leg but is not impinging on it. Adjust the back legs to be 2 or 3 notches above the front legs (i.e., the back legs of the Lift are taller than the front legs).

CAUTION: Ensure the Lift and Support Plate are covered completely by sterile drape during the procedure. Use towels as necessary to minimize direct contact between the patient and all surfaces of both the Lift and Support Plate.

15.5 Prepare the patient for invasive hemodynamic monitoring.

16.0 MITRACLIP™ G5 SYSTEM PREPARATION BEFORE USE

WARNING: DO NOT use the MitraClip™ G5 System after the "Use By" date stated on the package label, and never reuse or re-sterilize the system. Use of expired, reused, or re-sterilized devices may result in infection, malfunction of the device and other serious injury or death.

WARNING: Always inspect the MitraClip™ G5 System and its packaging to verify no damage has occurred as a result of shipping and handling and that the sterile barrier has not been compromised. DO NOT use the device if damage is detected. Use of product with a compromised sterile barrier may result in infection. Use of damaged product may result in patient injury.

WARNING: DO NOT handle the Implant directly; leave it in the protective cover to avoid potential contamination. Removal of the protective cover may result in infection. Removal of the protective cover may result in damaged product which may result in patient injury.

The preparation is most easily accomplished with the aid of an assistant.

16.1 MitraClip™ G5 Steerable Guide Catheter Preparation

WARNING: All lumens contain air when shipped. Use proper de-airing techniques before and during use to minimize the risk of air embolism.

16.1.1 Remove the Guide from sterile packaging and place in the sterile field.

16.1.2 Carefully remove the white Guide tip shape retainer and transparent protective tubing from the Guide tip.

16.1.3 Inspect Steerable Guide Catheter and Dilator to verify they are undamaged.

WARNING: DO NOT use if damage is detected. Use of damaged product may result in air embolism, vascular and / or cardiac injury.

16.1.4 Remove the Silicone Pad from the MitraClip™ G5 Steerable Guide Catheter tray.

16.1.5 Fill a basin with heparinized saline.

16.1.6 Flush and de-air the Guide and Dilator with heparinized saline:

16.1.6.1 Connect 3-way stopcocks to the Guide and Dilator flush ports, de-air the Dilator, and close the stopcock and the Rotating Hemostatic Valve.

16.1.6.2 Hydrate 5-10 cm of the distal end of the Dilator with heparinized saline and insert the Dilator approximately 10 cm into Guide then remove.

16.1.6.3 Connect high pressure tubing and a 50–60 cc syringe filled with heparinized saline to the Guide flush port.

16.1.6.4 De-air the Guide.

16.1.6.4.1 With the tip raised, displace all air from the Guide while tapping along the length of the catheter shaft.

16.1.6.4.2 Cover the Guide tip with finger once heparinized saline exits the Guide.

16.1.6.4.3 Close the Guide stopcock.

16.1.7 Submerge the Guide tip in the basin of heparinized saline.

16.1.8 While the Guide tip is submerged in the basin of heparinized saline, remove finger from Guide tip and check the Guide valve for leaks by raising the handle to a vertical position for a minimum of 30 seconds.

16.1.9 Hydrate 5-10 cm of the distal end of the Dilator with heparinized saline.

16.1.10 Cover the Guide tip with finger and insert the Dilator into the Guide while Guide tip remains submerged in the basin of heparinized saline.

16.1.10.1 While advancing the Dilator, continually watch for air in the Guide Hemostasis Valve housing. If needed, remove finger from Guide tip and aspirate while assuring the Guide tip is submerged.

16.1.11 Remove finger from tip of Guide when the Dilator tip approaches the Guide tip.

16.1.12 Position the Dilator to create a smooth transition.

16.2 Stabilizer Preparation

Inspect the Stabilizer to confirm it is undamaged and set the Stabilizer aside in a protected sterile environment for later use.

16.3 MitraClip™ G5 Delivery System Preparation

16.3.1 Confirm the DC Fastener is released and rotate the DC Handle 90 degrees clockwise when the CDS is removed from sterile packaging and place in the sterile field. Secure the DC Fastener.

16.3.2 Inspect all MitraClip™ G5 Delivery System parts, including the Implant, DC shaft, and Sleeve to verify they are undamaged. Ensure the Lock Line Release Latch is completely lowered.

WARNING: **DO NOT use the device if damage is detected. Use of damaged product may result in air embolism, Implant or device component embolization, vascular and / or cardiac injury.**

16.3.3 Connect a 3-way stopcock to the bottom DC flush port.

16.3.4 Remove the red cap from the Introducer and place the cap on the sterile table for later use.

16.3.5 Connect a 3-way stopcock to the Introducer flush port.

16.3.6 Attach a 50-60 cc syringe filled with heparinized saline to the 3-way stopcock on the Introducer, de-air the Introducer and close the stopcock.

16.3.7 Connect a high-pressure tube to the drip line from the pressurized bag with sterile heparinized saline; flush and de-air the line.

16.3.8 Ensure the pressure of the flush bag is set to 300 mmHg. Connect the high-pressure tube to the 3-way stopcock on the bottom flush port of the DC Handle.

16.3.9 Increase the flush rate to the CDS by fully opening the flush through the high-pressure tube. Flush and de-air DC Handle and all lumens of the DC with heparinized saline.

16.3.10 Release the DC Fastener. Retract the DC handle at a consistent, slow pace while maintaining control throughout the movement. Secure the DC Fastener.

16.3.11 Use the red cap to close off top flush port of the DC Handle.

16.3.12 Flush and de-air the Sleeve with heparinized saline.

16.3.13 Release the DC Fastener. Advance and retract the DC handle once at a consistent, slow pace while maintaining control throughout the movement to remove residual air from the lumen.

Note: Avoid fast and/or excessive DC Handle translations as that may create a vacuum and prolong the CDS de-airing process.

WARNING: DO NOT use excessive force when pulling the DC (Delivery Catheter) Radiopaque Ring against the Sleeve tip, while translating the DC shaft. It may result in device damage including distal tip embolization.

16.3.14 Advance the DC Handle at a consistent, slow pace while maintaining control throughout the movement. Secure the DC Fastener.

16.3.15 With the tip raised to a level above the DC Handle and the shaft held taut, displace all air from the DC and sleeve while tapping along the length of the catheter shaft.

16.3.16 Confirm continuous flow from the distal end of the DC and Sleeve.

Delivery Catheter and Implant Inspection

WARNING: DO NOT handle the Implant directly; leave in the protective cover to avoid potential contamination. Removal of the protective cover may result in infection. Removal of the protective cover may result in damaged product which may result in patient injury.

16.3.17 *Raise the Gripper(s)*, *Unlock the Implant*, and confirm the Implant is at *Grasping Arm Angle.*

NOTE: If *Implant Arm Angle* is greater than *Grasping Arm Angle*, *Close the Implant Arms* to *Grasping Arm Angle*; if *Implant Arm Angle* is less than *Grasping Arm Angle*, *Open the Implant Arms* to *Grasping Arm Angle*.

16.3.18 *Lower the Gripper(s)* once to de-air the lumens.

16.3.19 *Lock the Implant.*

16.3.20 *Close the Implant Arms* to approximately 20 degrees and *Test the Lock*.

WARNING: **DO NOT turn the Arm Positioner more than 1 turn in the "Open" direction from neutral. Failure to stop turning the Arm Positioner at 1 turn in the "Open" direction past neutral may result in Implant opening or device damage which could cause the Implant to become non-functional and lead to embolization and / or conversion to surgical intervention.**

16.3.21 Return the *Arm Positioner to Neutral.*

16.3.22 *Raise the Gripper(s)*, *Unlock the Implant*, and *Invert the Implant Arms*.

WARNING: **DO NOT over rotate the Lock Knob. Damage could occur causing the Implant to not lock, unlock, or open. Inability to open the Implant may result in valve injury or lead to deployment of the Implant in an unintended location.**

WARNING: **Turning the Arm Positioner in the "Open" direction more than 1 full turn past an *Implant Arm Angle* of 180° or turning past when resistance is first noted may result in device damage which could cause the Implant to become non-functional and lead to embolization, and / or conversion to surgical intervention.**

16.3.23 *Lock the Implant*, fully *Close the Implant Arms* and *Lower the Gripper(s).*

16.3.24 Release the DC Fastener. Retract the DC fully against the Sleeve in a consistent, slow pace while maintaining control of the movement, and then secure the DC Fastener.

NOTE: Avoid fast and/or excessive DC Handle translations as that may create a vacuum and prolong the CDS de-airing process.

WARNING: **Do not use excessive force when pulling the DC (Delivery Catheter) Radiopaque Ring against the Sleeve tip while translating the DC shaft. It may result in device damage including distal tip embolization.**

16.3.25 Without removing the protective cover, carefully slide the Introducer over the Implant and stop when the tip of the Implant is just proximal to the tip of the Introducer.

WARNING: **DO NOT compress the Implant Arms. Compressing the Implant Arms may result in inability to open the Implant. Inability to open the Implant may result in valve injury or lead to deployment of the Implant in an unintended location.**

WARNING: **Fully *Close the Implant Arms* before insertion or retraction into the Introducer. Failure to do so may result in difficulty or inability to advance or retract the Implant, which may result in vascular and / or cardiac injury, air embolism, and / or the need for surgical intervention.**

16.3.26 Return the *Arm Positioner to Neutral.*

16.3.27 Temporarily discontinue heparinized saline flush.

16.3.28 Re-start heparinized saline flushes just before the use of CDS.

WARNING:

- **Heparinized saline flush should be continuous throughout the procedure.**
- **Discontinuing flush may result in air embolism and / or thrombus formation.**
- **Ensure flow is visible through the drip chamber and that the tubing is free from kinks and / or obstruction.**
- **Ensure pressure of 300 mmHg is maintained.**

17.0 ACCESS TO THE MITRAL VALVE

NOTE: This is a suggested sequence for the procedure. Variations may be used based upon patient anatomy.

17.1 Access the LA to accommodate the Guide tip using transvenous, transseptal techniques and equipment.

17.2 Heparinize the patient.

WARNING: Failure to administer heparin once transseptal access has been achieved may result in thrombus formation.

17.3 Carefully place an exchange length supportive guidewire in the left upper pulmonary vein or LA. Dilate the subcutaneous tissue and femoral vein to accommodate the Guide shaft using standard dilation technique.

18.0 MITRACLIP™ G5 STEERABLE GUIDE CATHETER INSERTION

WARNING: Confirm a smooth transition between the Dilator and the tip of the Guide to minimize the risk of vascular and / or cardiac injury.

CAUTION: Always use pressure monitoring, echocardiography and fluoroscopy for guidance and observation during use of the MitraClip™ G5 System.

WARNING: Always use a careful, deliberate, and iterative approach to positioning the MitraClip™ G5 System. It is recommended to make multiple small adjustments rather than single large adjustments. Large adjustments may result in vascular and / or cardiac injury.

18.1 Rotate the +/- Knob in the "-" direction until the Guide curve is substantially straightened.

18.2 Wet the surface of the Guide shaft with sterile saline.

18.3 Insert the Guide-Dilator assembly over the stationary guidewire into the femoral vein.

WARNING: DO NOT use excessive force to advance or manipulate the Guide-Dilator assembly. If resistance is encountered, use echocardiography and / or fluoroscopy to assess before proceeding. Use of excessive force may result in arrhythmias, vascular and / or cardiac injury, including creation of a clinically significant atrial septal defect.

18.4 Advance the Guide-Dilator assembly to the RA. Rotate the +/- Knob to Neutral, then place tip of the Dilator partially across the atrial septum.

18.5 Slowly dilate the atrial septum by gradually advancing the tip of the Guide-Dilator assembly.

WARNING: DO NOT rapidly advance the Guide-Dilator assembly across the atrial septum. Rapid advancement may result in vascular and / or cardiac injury.

18.6 Advance the Guide-Dilator assembly until the tip of the Guide extends to the center of the LA.

18.7 Adjust Guide deflection and torque to position the tip away from adjacent tissues.

18.8 Place the Silicone Pad on the sterile drape over the Lift. Place the Stabilizer onto the Silicone Pad, such that the Stabilizer Guide Dock is underneath the Guide Attach.

18.9 Rotate the Guide Attach independently from the Guide as necessary to align the Guide Attach pins with the Guide Dock.

18.10 Insert the Guide Attach into the Guide Dock.

CAUTION: Ensure that the Stabilizer is secure on the Lift to prevent any undesirable movement.

18.11 Retract the Dilator approximately 5 cm into the Guide, leaving the guide wire in the left upper pulmonary vein or LA.

18.12 Retract the guidewire into the tip of the Dilator. Remove the Dilator and guidewire while gently aspirating the Guide (starting when the Dilator is approximately halfway retracted into the Guide, approximately 40 cm) using a 50–60 cc syringe. Cover Guide Hemostasis Valve with finger upon Dilator removal.

NOTE: Avoid contacting tissue or creating a vacuum in the Guide lumen. If necessary, position the Guide handle below the level of the LA to allow blood to fill the Guide lumen.

WARNING: DO NOT create a vacuum while removing the Dilator from the Guide; air may enter the lumen of the Guide which may result in air embolism.

WARNING: Failure to fully retract guidewire into the Dilator may result in air embolism.

19.0 MITRACLIP™ G5 DELIVERY SYSTEM INSERTION

19.1 Confirm the Guide lumen is completely de-aired.

WARNING: To minimize the potential of air embolism, DO NOT introduce the CDS into the Guide until the Guide lumen has been completely de-aired.

19.2 Confirm the CDS is completely de-aired and there is a slow, continuous heparinized saline flush through both the Sleeve and the DC and that the tip of the Implant is just proximal to the tip of the Introducer.

WARNING: Failure to continuously flush the CDS with heparinized saline may reduce device performance.

WARNING:

- **Heparinized saline flush should be continuous throughout the procedure.**
- **Discontinuing flush may result in air embolism and / or thrombus formation.**
- **Ensure flow is visible through the drip chamber and that tubing is free from kinks and / or obstruction.**
- **Ensure pressure of 300 mmHg is maintained.**

19.3 Carefully remove the protective cover surrounding the Implant and the Introducer.

19.4 Confirm that the stopcock on the Introducer flush port is closed and that the Introducer is de-aired.

19.5 While flushing heparinized saline on the Guide Hemostasis Valve, place the tip of the Introducer against the Guide Hemostasis Valve and advance the Introducer straight into the valve in a continuous motion while rotating the Introducer in small clockwise and counterclockwise motions until the Implant can be observed distal to the valve.

WARNING: **DO NOT continue to advance the Introducer if resistance is felt; the Guide Hemostasis Valve, Introducer or the Implant may be damaged. Damage to these components may result in air embolism, vascular or cardiac injury.**

WARNING: **To minimize the potential of air embolism, ensure proper de-airing when inserting the Introducer into the Guide Hemostasis Valve.**

19.6 Leave the Introducer fully inserted in the Guide Hemostasis Valve throughout the procedure.

19.7 Align the Longitudinal Alignment Marker on the Sleeve shaft with the Alignment Marker on the Guide Hemostasis Valve.

19.8 Turn the +/- Knob to neutral then carefully advance the CDS through the Guide under fluoroscopic guidance. Stop when tip of the Implant is even with the tip of the Guide.

NOTE: If resistance to CDS advancement is felt, reduce Guide deflection.

19.9 Under echocardiographic guidance, advance the CDS and retract the Guide iteratively as needed while maintaining the Guide in the LA. Stop when the Guide RO Tip Ring is between the RO Alignment Markers of the Sleeve, as confirmed under fluoroscopic guidance.

19.10 *Secure the DS Attach* and confirm the CDS is secured within the Stabilizer.

19.11 Confirm that the Implant is free from the left atrial wall and valve tissue.

WARNING: **Failure to confirm that the Implant is free from the left atrial wall and valve tissue may result in cardiac injury.**

20.0 INITIAL MITRACLIP™ G5 SYSTEM POSITIONING IN THE LEFT ATRIUM

NOTE: Positioning is achieved with iterative adjustments of the Guide and CDS using torque, translation, and knob adjustments. The goals of positioning are:

A. Positioning the Implant centrally over the valve with respect to anterior-posterior and medial-lateral directions.

B. Aligning the Implant so the DC Shaft is perpendicular to the plane of the mitral valve.

C. Positioning the distal tip of the Implant above the leaflets.

WARNING: Excessive torque on the Guide and translation of the MitraClip™ G5 System may inadvertently displace the tip of the Guide from the LA, which may result in arrhythmias or cardiac injury.

WARNING: DO NOT continue to rotate or manipulate any of the knobs if significant resistance is noted; device damage may occur and result in cardiac injury.

20.1 Adjust the Guide position as necessary to maintain that the Implant is free from adjacent tissue.

20.2 Adjust Sleeve deflection using the M/L Knob and / or the A/P Knob to deflect the Implant towards the apex. Retract the DC Radiopaque Ring against the Sleeve tip as necessary.

20.3 During Sleeve deflections confirm that the Guide RO Tip Ring is between the RO Alignment Markers of the Sleeve prior to making maximum Sleeve deflections.

WARNING: DO NOT deflect the Sleeve tip more than 90 degrees as device damage may occur. Use of damaged product may result in cardiac injury.

20.4 Confirm the CDS is secured within the Stabilizer.

20.5 To reposition the MitraClip™ G5 System, move the Stabilizer with the system until positioning is adequate.

20.6 Adjust the MitraClip™ G5 System position to maintain adequate height above the mitral valve in the LA.

WARNING: Maintain the Implant above the leaflets until ready to grasp to minimize the risk of Implant entanglement in the chordal apparatus. Implant entanglement may result in cardiac injury, worsening mitral regurgitation, difficulty, or inability to remove the Implant and conversion to surgical intervention.

21.0 FINAL MITRACLIP™ G5 SYSTEM POSITIONING

21.1 *Raise the Gripper(s).*

CAUTION: Raising the Grippers more often than needed or retracting the Gripper Levers forcefully may damage the Gripper cover or Gripper line and impair CDS performance.

21.2 *Unlock the Implant* and *Open the Implant Arms* to approximately 180 degrees.

WARNING: DO NOT over rotate the Lock Knob. Damage could occur causing the Implant to not lock, unlock, or open. Inability to open the Implant may result in valve injury or lead to deployment of the Implant in an unintended location.

21.3 Adjust the MitraClip™ G5 System to reposition the Implant as necessary. Confirm that the distal tip of the Implant is above the leaflets.

21.4 Release the DC Fastener. Rotate the DC handle to align the Implant Arms perpendicular to the line of coaptation. DO NOT rotate the Implant more than 90 degrees in each direction.

21.5 Carefully translate the DC shaft multiple times to release stored torque. Fully retract the DC. Secure the DC Fastener.

WARNING: Fully release stored torque. If not done, it may result in unwanted Implant Arm orientation changes during grasping. Torque of the DC Handle more than 180 degrees may result in DC damage and cardiac injury.

21.6 *Identify Gripper Orientation*.

21.7 Close the Implant to an *Implant Arm Angle* of approximately 60 degrees.

21.8 Complete final MitraClip™ G5 System positioning in the LA using multiple imaging planes.

22.0 GRASPING THE LEAFLETS AND VERIFYING THE GRASP

22.1 Advance the DC distally to position the Implant below the valve, until the leaflets are mobile above the Implant Arms. Ensure that the Implant Arms are oriented perpendicular to the line of coaptation.

WARNING: Do confirm that the Implant Arms are perpendicular to the line of coaptation. Failure to do so may result in loss of leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).

WARNING: DO NOT make substantial Implant Arm orientation adjustment in the LV. Implant entanglement in chordae may result in cardiac injury and worsening mitral regurgitation; and may result in difficulty or inability to remove the Implant, and conversion to surgical intervention.

WARNING: Always ensure that either the Grippers are raised or that the Implant is closed while in the LV to avoid potential cardiac injury.

22.2 *Open the Implant Arms* to the *Grasping Arm Angle*.

22.3 Without using excessive force, retract the DC to grasp the anterior and posterior leaflets.

WARNING: An improper grasp will allow one or both leaflets to move freely. Closing and deploying the Implant in this situation may result in loss of leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).

22.4 If the grasp appears satisfactory, *Lower the Gripper(s)* onto the leaflets.

NOTE: Simultaneous leaflet capture with both Grippers should be attempted first. If unsuccessful, *Raise the Gripper(s)* to release leaflet capture and *Lower the Gripper(s)* to capture leaflets. Raising and lowering Gripper(s) can be done simultaneously or independently.

WARNING: DO NOT advance the DC handle or adjust the position of the MitraClip™ G5 System in a way that increases tension on the leaflets after grasping the leaflets, as valve injury may occur.

22.4.1 If both Grippers have not lowered:

22.4.1.1 *Lock the Implant.*

22.4.1.2 Confirm both Grippers have lowered.

22.4.1.3 *Unlock the Implant.*

WARNING: Failure to confirm that both Grippers have been lowered onto the leaflets prior to closing the Implant may result in loss of leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).

22.5 Use echocardiographic imaging during grasping to verify proper leaflet insertion of both leaflets. Satisfactory grasp is confirmed by observation of:

- Leaflet immobilization
- Single or multiple valve orifice(s)
- Limited leaflet mobility relative to the tips of both Implant Arms

22.5.1 If grasping fails to hold both leaflets and the Implant retracts to the LA, reposition the MitraClip™ G5 System.

22.5.1.1 *Raise the Gripper(s)* and reorient the Implant Arms in the LA, as needed, then repeat grasping steps (refer to Section 21.0 and 22.0).

22.5.1.1.1 If significant repositioning is necessary, fully *Close the Implant Arms* and *Lower the Gripper(s)* then repeat positioning and grasping steps.

22.5.2 If the Sleeve limits DC travel during grasping, an inadequate grasp may require repositioning of the MitraClip™ G5 System.

22.5.2.1 *Raise the Gripper(s)* and *Open the Implant Arms* to approximately 180 degrees and advance the DC handle. Repeat positioning and grasping steps as necessary (refer to Section 21.0 and 22.0).

22.6 *Lock the Implant.*

WARNING: Failure to *Lock the Implant* may result in loss of leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).

23.0 CLOSING THE IMPLANT AND EVALUATING IMPLANT POSITION

23.1 Slowly *Close the Implant Arms* just until the leaflets are coapted and MR is sufficiently reduced. The Implant should maintain a distinct "V" shape. Ensure to release tension on the DC as the Implant is closed and secure the DC Fastener.

NOTE: Use echocardiographic imaging during closing of the Implant to verify proper leaflet insertion of both leaflets throughout the entire closure sequence.

WARNING: **DO NOT use excessive force to close the Implant further than is necessary to adequately reduce MR. Leaflet injury may occur. DO NOT close the Implant too tightly as it may result in leaflet injury or an inability to deploy the Implant. Inability to deploy the Implant may result in worsening mitral regurgitation, cardiac injury, a single leaflet device attachment (SLDA), and / or conversion to surgical intervention.**

WARNING: **Failure to turn the Arm Positioner at least ½ turn in the "Close" direction after locking the Implant may result in loss of leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).**

23.2 Use echocardiographic imaging to verify valve function, satisfactory coaptation, and insertion of both leaflets by observation of:

- Leaflet immobilization
- Single or multiple valve orifice(s)
- Limited leaflet mobility relative to the tips of both Implant Arms
- Adequate MR reduction

23.2.1 If the Implant position is not satisfactory, *Raise the Gripper(s), Unlock the Implant* and *Invert the Implant Arms*.

WARNING: **Turning the Arm Positioner in the "Open" direction more than 1 full turn past an *Implant Arm Angle* of 180° or turning past when resistance is first noted may result in device damage which could cause the Implant to become non-functional and lead to embolization, and / or conversion to surgical intervention.**

23.2.2 Retract the inverted Implant into the LA.

23.2.3 Confirm both leaflets move freely.

23.2.4 Repeat positioning steps, as necessary, then repeat grasping steps.

24.0 MITRACLIP™ G5 IMPLANT PRE-DEPLOYMENT IMPLANT ASSESSMENT

24.1 Confirm DC Handle is secure.

WARNING: Do secure the DC Handle. If not done, it may result in leaflet injury or loss of leaflet insertion with resultant worsening mitral regurgitation, single leaflet device attachment (SLDA), and / or conversion to surgical intervention.

24.2 *Test the Lock*.

WARNING: DO NOT turn the Arm Positioner more than 1 turn in the "Open" direction from neutral. Failure to stop turning the Arm Positioner at 1 turn in the "Open" direction past neutral may result in Implant opening or device damage which could cause the Implant to become non-functional and lead to embolization and / or conversion to surgical intervention.

24.3 Turn the Arm Positioner to the "closed" side of the neutral position.

24.4 Perform mean pressure gradient assessment prior to proceeding to deployment.

25.0 IMPLANT DEPLOYMENT

25.1 Deployment Step 1: Lock Line Removal

25.1.1 Increase the flush rate to the CDS Handle.

25.1.2 *Lift the Lock Line Release Latch*

WARNING: Once the latch is lifted, be careful to not inadvertently lower the latch prior to Lock Line removal. DO NOT Lift the Lock Line Release Latch until deployment. Premature lifting of the Lock Line Release Latch may result in the inability to *Unlock the Implant*.

25.1.3 *Remove the Lock Knob*.

WARNING: If resistance is felt during Lock Knob removal, stop and confirm that the Lock Line Release Latch is fully lifted to the hard stop before continuing with Lock Knob removal.

25.1.4 *Test the Lock*.

NOTE: The Implant Arms may open slightly (~5°) and then remain in a stable position. If Arms open more than slightly, *Close the Implant Arms* to the desired Arm position and re-*Test the Lock*.

WARNING: DO NOT turn the Arm Positioner more than 1 turn in the "Open" direction from neutral. Failure to stop turning the Arm Positioner at 1 turn in the "Open" direction past neutral may result in Implant opening or device damage which could cause the Implant to become non-functional and lead to embolization and / or conversion to surgical intervention.

25.1.5 Turn the *Arm Positioner to Neutral*.

25.2 Deployment Step 2: Delivery Catheter Shaft Detachment

25.2.1 Confirm that the Arm Positioner is Neutral. Remove the Release Pin from the DC Handle.

25.2.2 Turn the Arm Positioner in the “Open” direction until the hard stop.

NOTE: After the Release Pin is removed, turning the Arm Positioner in the “Open” direction will not open the Implant Arms.

25.2.3 Turn the Actuator Knob of the DC approximately 8 turns in the direction of the arrow printed on the Actuator Knob.

If it is difficult to turn the Actuator Knob, STOP and confirm that the Arm Positioner has been turned in the “Open” direction until the hard stop.

WARNING: Stop turning the Actuator Knob when resistance is felt, otherwise it may result in inability to deploy the Implant. Inability to deploy the Implant may result in worsening mitral regurgitation, cardiac injury, a single leaflet device attachment (SLDA), and / or conversion to surgical intervention.

25.2.4 Retract the Actuator Knob approximately 0.5 cm after it is fully unthreaded.

25.2.5 Fully retract the Gripper Levers.

WARNING: Do retract the Gripper Levers fully prior to retracting the DC Handle. Failure to retract the Gripper Levers may result in higher forces to deploy the Implant. This may result in worsening mitral regurgitation, cardiac injury, or a single leaflet device attachment (SLDA).

25.2.6 Release the DC Fastener, and slowly retract the DC Handle until the DC Radiopaque Ring is against the tip of the Sleeve.

25.2.6.1 If resistance is felt during DC Detachment and Implant separation from the DC can be confirmed by fluoroscopy, confirm the Gripper Levers are fully retracted. If resistance is still felt, access the Gripper Lines.

25.2.6.2 If resistance is felt during DC Detachment and Implant separation from the DC cannot be confirmed by fluoroscopy:

25.2.6.2.1 Confirm that the Actuator Knob is retracted approximately 0.5 cm beyond the fully exposed Release Pin Groove and the Gripper Levers are fully retracted.

25.2.6.2.2 If resistance is still felt during DC Detachment, confirm by fluoroscopy that the DC Shaft and the Implant are coaxially aligned. If they are not aligned, secure the DC fastener and slowly translate the Stabilizer and / or rotate the Guide until the angulation between the DC Shaft and the Implant is reduced or eliminated (as observed under fluoroscopic imaging).

25.2.6.2.3 If resistance is still felt, release the DC Fastener, retract the Actuator Knob an additional 0.5 cm.

25.2.6.2.4 If resistance is still felt, secure the DC Fastener and fully advance both Gripper Levers.

25.2.6.2.5 If resistance is still felt, access the Gripper Lines.

25.2.6.2.6 If resistance is still felt, secure the DC Fastener and repeat alignment steps to reduce / eliminate the angulation between the DC shaft and the Implant (as observed under fluoroscopic imaging).

25.2.7 Confirm that the DC Fastener is secure.

25.2.8 **Confirm that the Implant position is stable.**

25.2.9 Use echocardiographic imaging to verify valve function, satisfactory coaptation, and insertion of both leaflets by observation of:

- Leaflet immobilization
- Single or multiple valve orifice(s)
- Limited leaflet mobility relative to the tips of both Implant Arms
- Adequate MR reduction

25.2.10 **If placing an additional Implant proceed to Section 26.0. If not placing an additional Implant proceed to Section 27.0.**

26.0 ADDITIONAL MITRACLIP™ G5 IMPLANT PLACEMENT

WARNING: Use caution not to displace or dislodge an implanted Implant when placing an additional Implant; Implant detachment from leaflet(s) may occur which may result in a single leaflet device attachment (SLDA) or device embolization.

26.1 When placing an additional Implant, the following are recommended:

26.1.1 In the LA, ensure Implant Arms are oriented perpendicular to the line of coaptation and Grippers are raised.

26.1.2 Use both fluoroscopy and echocardiography when crossing into the LV and during grasping.

26.1.3 Cross into the LV with an *Implant Arm Angle* of < 60 degrees.

WARNING: DO NOT use excessive force or retraction distance during grasping. This may compromise leaflet capture and insertion. Loss of leaflet capture and insertion may result in inadequate reduction of mitral regurgitation and may result in a single leaflet device attachment (SLDA).

27.0 MITRACLIP™ G5 SYSTEM REMOVAL

WARNING: When removing the MitraClip™ G5 Delivery System while leaving the Guide in place, always retract the CDS by pulling only on the Sleeve Handle. Retracting the CDS by pulling on the DC Handle may result in device damage and / or device or component embolization, and may result in vascular and / or cardiac injury.

WARNING: Do release the DC Fastener before releasing Sleeve curves otherwise it may result in device damage and / or device or component embolization.

WARNING: Use echocardiographic guidance while releasing Sleeve deflection. Failure to do so may result in cardiac injury.

27.1 MitraClip™ G5 System Removal After Implant Deployment

27.1.1 **Removal of the CDS While Leaving the Guide in Place.**

27.1.1.1 Release the DC Fastener.

27.1.1.2 Slowly release Sleeve deflection by rotating the M/L Knob and the A/P Knob to neutral.

27.1.1.3 Secure DC Fastener once Sleeve curves are released.

27.1.1.4 Straighten the Guide with the +/- Knob when the Delivery Catheter tip is free from the left atrial wall and the mitral valve.

27.1.1.5 *Release the DS Attach* and retract the CDS approximately 10 cm into the Guide by pulling only on the Sleeve Handle.

27.1.1.6 Confirm that the Introducer is still fully advanced in the Guide Hemostasis Valve.

27.1.1.7 Retract the CDS by pulling only on the Sleeve Handle and position the Delivery Catheter tip inside the Introducer. Begin gently aspirating the Guide (starting when the CDS is approximately halfway into the Guide, approximately 40 cm retracted) using a 50–60 cc syringe.

27.1.1.8 Remove the CDS and the Introducer simultaneously from the Guide by pulling on the Sleeve shaft and Introducer. Ensure the Delivery Catheter tip is inside the Introducer by visualizing the Proximal Sleeve alignment marker just outside the Introducer. Aspirate the Guide during removal of the CDS and Introducer. Cover Guide Hemostasis Valve with finger upon CDS removal. If necessary, position the Guide Handle below the level of the LA to allow blood to fill the Guide Lumen.

WARNING: DO NOT remove the tip of the CDS from the Guide without removing the Introducer simultaneously. Failure to remove the Introducer simultaneously may result in air embolism.

WARNING: DO NOT create a vacuum while removing the CDS from the Guide; air may enter the lumen of the Guide which may result in air embolism.

27.1.1.9 Aspirate using a 50–60 cc syringe to remove any remaining air from the Guide.

27.1.2 Removal of the CDS and Guide simultaneously.

27.1.2.1 Release the DC Fastener.

27.1.2.2 Slowly release Sleeve curves by rotating the M/L Knob and the A/P Knob to neutral.

27.1.2.3 Secure the DC Fastener once Sleeve curves are released.

27.1.2.4 Straighten the Guide with the +/- Knob when the Delivery Catheter tip is free from the left atrial wall and the mitral valve.

27.1.2.5 *Release the DS Attach* and retract the CDS approximately 10 cm into the Guide by pulling only on the Sleeve Handle.

27.1.2.6 Carefully retract the Guide tip into the RA. The Guide may be straightened further with the +/- Knob if desired.

27.1.2.7 Remove the MitraClip™ G5 System from the femoral vein, while providing hemostasis.

27.2 MitraClip™ G5 System Removal With Implant Attached

27.2.1 **Removal of the CDS while leaving the Guide in place.**

27.2.1.1 Confirm Implant is locked.

27.2.1.2 Fully *Close the Implant Arms* and continue to turn the Arm Positioner until it is no longer possible to rotate the Arm Positioner. Return the *Arm Positioner to Neutral*.

WARNING: Failure to follow Step 27.2.1.2 prior to retraction into the Guide may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.

27.2.1.3 *Lower the Gripper(s)*.

27.2.1.4 Release the DC Fastener and retract the DC Handle until the DC Radiopaque Ring is fully against the tip of the Sleeve.

27.2.1.5 Slowly release Sleeve deflection by rotating the M/L Knob and the A/P Knob to neutral.

27.2.1.6 Rotate DC handle such that the Implant Arms are perpendicular to the guide curve plane.

27.2.1.7 Secure the DC Fastener once Sleeve curves are released.

27.2.1.8 Straighten the Guide with the +/- Knob when the tip of the MitraClip™ G5 Implant is free from the left atrial wall and the mitral valve.

WARNING: Straighten the Guide prior to retracting the Implant into the Guide. If not done, it may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.

27.2.1.9 *Release the DS Attach* and retract the CDS into the Guide by pulling only on the Sleeve Handle.

NOTE: If resistance is noted, advance and rotate the Implant by rotating the DC Handle then retract the CDS into the Guide. The Guide and / or Sleeve position may also be adjusted to facilitate Implant entry into the Guide. If necessary, retract the Sleeve or advance the Implant to create a 2-3cm separation to facilitate Implant entry into the Guide.

WARNING: **Use fluoroscopic guidance while retracting the CDS into the Guide. Failure to do so may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.**

27.2.1.10 Confirm that the Introducer is still fully advanced in the Guide Hemostasis Valve.

27.2.1.11 Retract the CDS by pulling only on the Sleeve Handle and position the Implant inside the Introducer. Begin gently aspirating the Guide (starting when the CDS is approximately halfway into the Guide, approximately 40 cm retracted) using a 50–60 cc syringe.

27.2.1.12 Remove CDS and Introducer simultaneously from the Guide by pulling on the Sleeve shaft and Introducer. Ensure the Implant is inside the Introducer by visualizing the Proximal Sleeve alignment marker just outside the Introducer. Aspirate the Guide during removal of the CDS and Introducer. If necessary, position the Guide Handle below the level of the LA to allow blood to fill the Guide lumen.

WARNING: **DO NOT remove the tip of the CDS from the Guide without removing the Introducer simultaneously and with the Implant inside the Introducer. Failure to remove the Introducer simultaneously may result in air embolism**.

WARNING: **DO NOT create a vacuum while removing the CDS from the Guide; air may enter the lumen of the Guide which may result in air embolism.**

WARNING: **DO NOT re-use the CDS after removal. Replace the CDS with a new device. Reinserting the CDS after removal may result in inability to open the Implant. Inability to open the Implant may result in valve injury or lead to deployment of the Implant in an unintended location.**

27.2.1.13 Aspirate using a 50–60 cc syringe to remove any remaining air from the Guide.

27.2.2 **<u>Simultaneous removal of CDS and Guide.</u>**

27.2.2.1 Confirm Implant is locked.

27.2.2.2 Fully *Close the Implant Arms* and continue to turn the Arm Positioner until it is no longer possible to rotate the Arm Positioner. Return the *Arm Positioner to Neutral*.

WARNING: **Failure to follow Step 27.2.2.2 prior to retraction into the Guide may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.**

27.2.2.3 *Lower the Gripper(s)*.

27.2.2.4 Release the DC Fastener and retract the DC Handle until the DC Radiopaque Ring is fully against the tip of the Sleeve.

27.2.2.5 Slowly release Sleeve deflection by rotating the M/L Knob and the A/P Knob to neutral.

27.2.2.6 Rotate DC handle such that the Implant Arms are perpendicular to the guide curve plane.

27.2.2.7 Secure the DC Fastener once Sleeve curves are released.

27.2.2.8 Straighten the Guide with the +/- Knob when the tip of the MitraClip™ G5 Implant is free from the left atrial wall and the mitral valve.

WARNING: **Straighten the Guide prior to retracting the Implant into the Guide. Failure to do so may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.**

27.2.2.9 *Release the DS Attach* and retract the CDS approximately 10 cm into the Guide by pulling only on the Sleeve Handle.

NOTE: If resistance is noted, advance and rotate the Implant by rotating the DC Handle then retract the CDS into the Guide. The Guide and / or Sleeve position may also be adjusted to facilitate Implant entry into the Guide. If necessary, retract the Sleeve or advance the Implant to create a 2-3 cm separation to facilitate Implant entry into the Guide.

WARNING: **Use fluoroscopic guidance while retracting the CDS into the Guide. Failure to do so may result in device damage, inability to remove the CDS and / or vascular and cardiac injury.**

27.2.2.10 Carefully retract the Guide tip into the RA. The Guide may be straightened further with the +/- Knob if desired.

27.2.2.11 Remove the MitraClip™ G5 System from the femoral vein, while providing hemostasis.

28.0 DISPOSAL

After use, these devices and packaging should be appropriately classified for disposal, e.g., biohazard, sharps, non-hazardous waste etc., and carefully disposed of in compliance with facility procedures and applicable laws and regulations.

29.0 CLINICAL DATA

29.1 Overview Of Clinical Studies

The MitraClip System was the first-in-class transcatheter therapy for the treatment of mitral regurgitation and is supported by numerous clinical studies that assessed the safety and efficacy of MitraClip in patients with MR. **Table 2** presents an overview of the prospective, multi-center pre-market studies in the MitraClip™ clinical program sponsored by Abbott in the United States including study design, sample size, enrollment criteria, follow-up schedule, and endpoints.

Following approval of the MitraClip System, several post-approval surveillance studies were initiated to confirm and evaluate the real-world safety and effectiveness of the MitraClip System. **Table 3** presents an overview of the post-approval and real-world surveillance studies in the MitraClip clinical program sponsored by Abbott in the U.S including study design, sample size, enrollment criteria, follow-up schedule, and endpoints.

Study results are described in Section 29.2.

Table 2: Overview of MitraClip™ US Pre-Market Clinical Studies

Study Design	Study	# of Subjects	# of Sites	Enroll-ment Period	Key Inclusion Criteria	Key Exclusion Criteria	Follow-up	Endpoint
Feasibility	EVEREST I	55	11	2003-2006	• MR≥3+ • Symptomatic or asymptomatic with: ○ LVEF 30-50% and / or LVESD 50-55mm or ○ LVEF 50-60% and LVESD <45 mm or ○ LVEF>60 and LVESD 45-55 mm • Candidate for mitral valve surgery including cardiopulmonary bypass	• LVEF<30%, and / or LVESD >55mm • MV orifice area <4.0 cm^2 • Leaflet anatomy which may preclude MitraClip™ device implantation, proper MitraClip™ device positioning on the leaflets or sufficient reduction in MR	Baseline, discharge, 30 days, 6, 12, 18 and 24 months, and 3, 4, and 5 years	Primary: Major Adverse Event rate through 30 days
Randomized Controlled Trial (RCT)	EVEREST II RCT	178[a] device (60 roll-ins) 80[a] surgical control	37	2005-2008	• MR≥3+ • Symptomatic with LVEF > 25% and LVESD ≤ 55 mm or asymptomatic with[a]: ○LVEF 25% to 60% ○LVESD ≥ 40 mm ○New onset of atrial fibrillation ○PASP>50mmHg at rest of >60 mmHg with exercise	• LVEF≤25%, and / or LVESD >55mm • MV orifice area <4.0 cm^2 • Leaflet anatomy which may preclude MitraClip™ device implantation, proper MitraClip™ device positioning on the leaflets or sufficient reduction in MR	Baseline, discharge, 30 days, 6, 12, 18 and 24 months, and 3, 4, and 5 years	• Primary Safety: Major Adverse Event rate through 30 days or discharge, whichever is greater • Primary Effectiveness: Freedom from death, MV surgery (for Device group) or re-operation (for Control group), and MR >2+ at 12 months • Secondary Effectiveness: ○Measures of LV Function ○SF-36 quality of life ○NYHA Functional Class

Study Design	Study	# of Subjects	# of Sites	Enrollment Period	Key Inclusion Criteria	Key Exclusion Criteria	Follow-up	Endpoint
Single-Arm Registry	EVEREST II High Risk Registry	78	25	2007-2008	• MR≥3+ • Predicted procedural mortality risk calculated using the STS surgical risk calculator of ≥ 12% or in the judgment of a cardiac surgeon the patient is considered a high risk surgical candidate due to the presence of one of the following indications: 1. Porcelain aorta, mobile ascending aortic atheroma 2. Post-radiation mediastinum 3. Previous mediastinitis 4. Functional MR with EF<40 5. Over 75 years old with EF<40 6. Re-operation with patent grafts 7. Two or more prior chest surgeries 8. Hepatic cirrhosis 9. Three or more of the following STS high risk factors 9.1 Creatinine > 2.5 mg/dL 9.2 Prior chest surgery 9.3 Age over 75 9.4 EF<35	• LVEF<20% and / or LVESD>60mm • Mitral valve orifice area <4.0 cm^2 • Leaflet anatomy which may preclude MitraClip™ device implantation, proper MitraClip™ device positioning on the leaflets or sufficient reduction in MR	Baseline, discharge, 30 days, 6, 12, 18 and 24 months, and 3, 4, and 5 years	• Primary Safety: Procedural mortality at 30 days • Major Secondary: ○Measures of LV Function ○SF-36 quality of life ○NYHA Functional Class ○Congestive Heart Failure Hospitalizations • Secondary Safety: ○Major Adverse Event rate at 30 days and 12 months
Continued Access Registry	REALISM High Risk	628	39	2009-2013	Same as EVEREST II High Risk Registry with the exception of the requirement for predicted procedural mortality risk ≥ 12%	Same as EVEREST II High Risk Registry	Baseline, discharge, 30 days, 6, 12, 18, and 24 months, and 3, 4, and 5 years	Same as High Risk Registry
	REALISM Non-High Risk	271	39	2009-2011	Same as EVEREST II RCT	Same as EVEREST II RCT	Baseline, discharge, 30 days, 6, 12, 18, and 24 months, and 3, 4, and 5 years	• Same as RCT • 6 Minute Walk Test (6MWT) Distance[b]

Study Design	Study	# of Subjects	# of Sites	Enroll-ment Period	Key Inclusion Criteria	Key Exclusion Criteria	Follow-up	Endpoint
Randomized Controlled Trial	COAPT	302 device +GDMT (51 roll-ins), 312 GDMT control	84	2012-2017	• Ischemic or non-ischemic cardiomyopathy with LVEF 20%-50% and LVESD ≤70 mm • Moderate-to-severe (3+) or severe (4+) secondary MR confirmed by an independent echo core laboratory prior to enrollment • NYHA Functional Class II, III or ambulatory IV despite stable maximally-tolerated GDMT regimen and cardiac resynchronization therapy (if appropriate) • At least one hospitalization for heart failure in the 12 months prior to subject registration and/or a corrected BNP ≥300 pg/ml or corrected NT-proBNP ≥1500 pg/ml measured within 90 days prior to subject registration • Surgery per local heart team assessment • (LVESD) is ≤70 mm assessed by site based on a transthoracic echocardiographic • Treating interventionalist believes secondary MR can be successfully treated with MitraClip™	• ACC/AHA stage D heart failure, hemodynamic instability or cardiogenic shock • Untreated clinically significant coronary artery disease requiring revascularization • COPD requiring continuous home oxygen therapy or chronic oral steroid use • Severe pulmonary hypertension or moderate or severe right ventricular dysfunction • Aortic or tricuspid valve disease requiring surgery or transcatheter intervention • MV orifice area <4.0 cm^2 • Life expectancy < 12 months due to non-cardiac reasons	Baseline, discharge, 30 days, 6, 12, 18, and 24 months, and 3, 4, and 5 years	• Primary Safety: Composite of device-related complications at 12 months • Primary Effectiveness: Recurrent heart failure hospitalizations through 24 months

[a] Of the 184 patients randomized to Device, 178 received Device. Of the 95 patients randomized to Control, 80 underwent mitral valve surgery.
[b] In protocol version dated November 17, 2008, only patients with NYHA Functional Class III or IV in the Non-High Risk arm were considered for a 6-minute walk test. In the amended protocol version dated September 14, 2010, all patients enrolled in REALISM are required to perform the 6-minute walk test.
6MWT=6 Minute Walk Test, ACC=American College of Cardiology, AHA=American Heart Association, COPD=Chronic Obstructive Pulmonary Disease, GDMT=Guideline Directed Medical Therapy, EF=Ejection fraction, LVEF=Left Ventricular Ejection Fraction, LVESD=Left Ventricular End-Systolic Dimension, MR=Mitral Regurgitation, MV=Mitral Valve, NYHA=New York Heart Association, PASP=Pulmonary Artery Systolic Pressure, SF-36=Short Form health survey questionnaire, STS=Society of Thoracic Surgeons

Table 3: Overview of Post-Approval or Real-World MitraClip™ US Clinical Studies

Study Design	Study	# of Subjects	# of Sites	Enroll-ment Period	Study Population	Data Source	Follow-up	Endpoint
Post-Approval Study	MitraClip Analysis Cohort	554	50	2013-2015	Subset of data from TVT registry from sites/centers of excellence experienced with performing / collecting 6MWT and KCCQ data.	Subset of the first 2000 MitraClip™ patients consecutively entered into the STS/ACC TVT Registry	Baseline, discharge, 30 days, 12 months	• Primary Safety: Freedom from a composite of death and device-related complications at 30 days post-procedure > performance goal of 80% • Primary Effectiveness: Confirm that patients achieve improvement in distance walked on the 6MWT at 12 months

Study Design	Study	# of Subjects	# of Sites	Enroll-ment Period	Study Population	Data Source	Follow-up	Endpoint
Real-World Surveillance	EXPAND	1041	58	2018-2019	Consented subjects treated commercially in the U.S or EU with the MitraClip™ NTR and MitraClip™ XTR Systems.	All data collected were monitored by qualified Abbott personnel. Major adverse events were adjudicated by an independent clinical event committee. Leaflet adverse events were adjudicated by an independent physician committee. Echocardiographic measurements were assessed by an independent echocardiographic core lab.	Baseline, discharge, 30 days, 6 and 12 months.	Descriptive Endpoints include: • Safety: Major adverse events at 30 days defined as a composite of all-cause death, myocardial infarction, stroke, or non-elective cardiovascular surgery for device related complications. • Performance: MR reduction to ≤2+ at 30 days
	EXPAND G4	1164 (101 in Phase I 1063 in Phase II)	60	2020-2022	Consented subjects treated commercially in the U.S, EU, Canada, Middle East, and Japan with the MitraClip™ G4 System.	All data collected was monitored by qualified Abbott personnel. Major adverse events were assessed by the site and reviewed by qualified Abbott safety personnel. Leaflet adverse events and echocardiographic measurements were assessed by an independent echocardiographic core lab.	Phase I: Baseline, discharge, 30 days Phase II: Baseline, discharge, 30 days, 6 months, 1, 2, 3, 4, and 5 years	Descriptive Endpoints include: • Safety: Major adverse events at 30 days defined as a composite of all-cause death, myocardial infarction, stroke, or non-elective cardiovascular surgery for device related complications. • Performance: MR reduction to ≤2+ at 30 days

6MWT=6 Minute Walk Test, ACC=American College of Cardiology, KCCQ=Kansas City Cardiomyopathy Questionnaire, MR=Mitral Regurgitation, STS=Society of Thoracic Surgeons, TVT=Transcatheter Valve Therapies

29.1.1 EVEREST I Trial (Feasibility)

The EVEREST I trial was a prospective, multi-center, registry trial designed to evaluate the preliminary safety and effectiveness of the MitraClip™ device in the treatment of moderate-to-severe (3+) or severe (4+) chronic MR using up to 2 MitraClip™ devices per patient (**Table 2**). The EVEREST I trial demonstrated the preliminary safety and feasibility of the MitraClip™ device as a percutaneous method for the reduction of MR severity. The study is now closed.

29.1.2 EVEREST II Randomized Clinical Trial (RCT)

The EVEREST II RCT was a landmark trial, being the first randomized trial to compare a percutaneous intervention for the reduction of MR to standard of care mitral valve surgery. The EVEREST II RCT was a prospective, blinded, randomized, controlled, multi-center study of 279 patients (184 MitraClip™, 95 surgical control) comparing the safety and effectiveness of the MitraClip™ to the standard of care mitral valve surgery. The intended population was patients with significant symptomatic mitral regurgitation (MR ≥ 3+) of either secondary MR or primary MR etiology that were non-high-risk candidates indicated for and who could undergo mitral valve surgery. Study design elements including key inclusion/exclusion criteria and endpoints are provided in **Table 2.** Patients were evaluated at baseline, discharge, 30 days, 6, 12, 18 and 24 months, and annually thereafter through 5 years. Results of this study showed that the safety advantages of the percutaneous procedure were offset by the diminution of MR reduction with MitraClip™ compared to surgery, and therefore good surgical candidates should continue to receive surgical intervention.

29.1.3 EVEREST II High Risk Registry and EVEREST II REALISM Continued Access Study - High Risk (EVEREST II HRR and REALISM HR)

The EVEREST II High Risk Registry (HRR) was a prospective, multi-center, registry designed to be adjunctive to the RCT and to evaluate the safety and effectiveness of the MitraClip™ device in the treatment of high surgical risk (≥ 12%) patients with moderate-to-severe (3+) or severe (4+) chronic MR using up to 2 MitraClip™ devices per patient (**Table 2**). The primary safety endpoint of the EVEREST II HRR was procedural mortality at 30 days or prior to discharge, whichever is longer. REALISM HR was a single-arm, self-controlled adjunctive study enrolling the same patient population as the EVEREST II HRR and designed to continue to collect safety and effectiveness data and allow patients continued access to the MitraClip™ during review of the PMA application.

29.1.4 COAPT (Cardiovascular Outcomes Assessment of the MitraClip™ Percutaneous Therapy for Heart Failure Patients with Functional Mitral Regurgitation)

The COAPT Trial was a prospective, randomized (1:1; MitraClip™ + guideline directed medical therapy (GDMT) vs. GDMT alone), open-label, multicenter investigational study intended to demonstrate: (1) MitraClip™ was safe in subjects with secondary MR, and (2) MitraClip could reduce recurrent HF hospitalization as compared to the GDMT Control group. The randomization was further stratified by study site and by cardiomyopathy etiology (e.g., ischemic or non-ischemic).

The COAPT Trial was conducted under the oversight of several independent committees, including: (1) a Steering Committee, which provided scientific and medical input on trial design, data collection, data analyses, and interpretation of results; (2) an independent Eligibility Committee, which confirmed that each subject was on optimal therapy including GDMT prior to being considered for the trial and that the subject was not appropriate for mitral valve surgery, even if randomized to the Control group; (3) a Central Echocardiography Core Laboratory (ECL), which was responsible for reviewing subject's screening echocardiography images to determine if the subject met the MR severity eligibility criterion prior to the subject being considered eligible for the trial, and for assessing MR severity and left ventricular measurements, along with other measures, at baseline and follow-ups; (4) a Clinical Events Committee (CEC), which adjudicated all adverse events per pre-established definitions (blinding was maintained whenever feasible); (5) a Data Monitoring Committee (DMC), which monitored the safety of subjects throughout the trial; and (6) a Contract Research Organization, which participated in source data verification. Details on the study design are summarized in **Table 2**. Additional details on study endpoints have been published previously [5].

29.1.5 Post-Approval Study – MitraClip Analysis Cohort

The MitraClip Post-Approval study was a prospective, multi-center, study that utilized a MitraClip Analysis Cohort (MAC) following approval of the MitraClip System in 2013 (**Table 3**). A subset of patients from the Transcatheter Valve Therapy (TVT) Registry treated with the MitraClip System were used in the MAC. The primary safety objective of the MAC was to demonstrate freedom from a composite of death and device-related complications at 30 days post-procedure was greater than a performance goal of 80%. The primary effectiveness objective was to confirm that patients achieved improvement in distance walked in the 6 Minute Walk Test (6MWT) at 12 months over baseline. Results from the MAC confirmed the safety and effectiveness of MitraClip in a real-world setting.

29.1.6 Real-World Use Surveillance Study – EXPAND

The EXPAND study was a prospective, multi-center, single-arm, international, post-market, observational study designed to collect real-world data on the use of the next generation MitraClip NTR and MitraClip XTR Systems (**Table 3**). Consented subjects commercially treated with the MitraClip NTR/XTR System in the U.S or Europe were considered. Safety was assessed via the evaluation of major adverse events (MAEs) at 30 days defined as a composite of all-cause mortality, myocardial infarction, stroke, or non-elective cardiovascular surgery for device-related complications. Effectiveness was evaluated as MR reduction to ≤2+ at 30 days. The 30-day findings in EXPAND demonstrated safe and effective treatment of patients with MR using the MitraClip NTR and XTR System.

29.1.7 Real-World Use Surveillance Study – EXPAND G4

The EXPAND G4 study is a prospective, multi-center, single-arm, international, post-market, observational study designed to collect real-world data on the use of the next generation MitraClip G4 System (**Table 3**). Consented subjects commercially treated with the MitraClip G4 System in the U.S, Europe, Canada, Israel, Saudi Arabia, and Japan were considered. Safety was assessed via the evaluation of major adverse events (MAEs) at 30 days defined as a composite of all-cause mortality, myocardial infarction, stroke, or non-elective cardiovascular surgery for device-related complications. Effectiveness was evaluated as MR reduction to ≤2+ at 30 days. The 30-day findings in EXPAND G4 demonstrated safe and effective treatment of patients with MR using the MitraClip G4 System.

29.2 Clinical Study Results

29.2.1 Prohibitive Risk Primary MR Patient Clinical Results

Data on 127 patients with significant symptomatic mitral regurgitation due to primary abnormality of the mitral apparatus (primary mitral regurgitation) determined to be at prohibitive risk for mitral valve surgery (Prohibitive Risk primary MR Cohort or PR primary MR Cohort) that were collected from the EVEREST II HRR and REALISM HR studies are provided in detail below. The analysis cohort of 127 subjects was developed post-hoc; this severely limits the statistical interpretability of reported data. These data were determined to adequately establish the safety, effectiveness, and positive benefit-risk profile of the MitraClip™ for the indicated population (PR primary MR) and are the basis for PMA approval. Additional details of the prohibitive risk PMR patients have been published previously.[1]-[4]

Prohibitive Risk primary MR patients treated with the MitraClip™ were elderly with a high rate of serious comorbidities (**Table 4**).

Table 4: Prohibitive Risk Primary MR MitraClip™ Cohort – Key Baseline Characteristics

Baseline Characteristic[a]	Prohibitive Risk Primary MR MitraClip™ Patients % (n/N) (N = 127)
Age (years)	82.4±8.7 (127)
Patients over 75 years of age	83.5% (106/127)
Female Gender	44.9% (57/127)
Body Mass Index (kg/m^2)	25.0±5.7 (127)
Coronary Artery Disease	72.8% (91/125)
Prior Myocardial Infarction	24.4% (31/127)
Atrial Fibrillation History	70.5% (86/122)
Prior Stroke	10.2% (13/127)
Diabetes	29.9% (38/127)
Moderate to Severe Renal Disease	28.3% (36/127)
Cardiomyopathy	23.6% (30/127)
Chronic Obstructive Pulmonary Disease (w/ or w/o Home O2)	31.5% (40/127)
Hypertension	88.2% (112/127)
Previous Cardiovascular Surgery	48.0% (61/127)
Previous Percutaneous Coronary Intervention	33.3% (42/126)
NYHA Functional Class III/IV Heart Failure	86.6% (110/127)
LV Ejection Fraction (%)	60.6±9.5 (112)
LV Internal Diameter, systole (cm)	3.4±0.8 (113)
STS Mortality Risk (determined at enrollment for replacement)[b], Mean±SD (N)	13.6±7.9 (127)

[a] Sample sizes or denominators smaller than the N reported for the group reflect missing data.
[b] STS replacement score calculated using the version of the calculator at the time of enrollment.
Values presented as Mean ± SD or % (n/N).

The mean Procedure Time, defined as the start time of the transseptal procedure to the time the Steerable Guide Catheter is removed, was approximately 2.5 hours (**Table 5**). Device time, defined as the time of insertion of the Steerable Guide Catheter to the time the MitraClip™ Delivery Catheter is retracted into the Steerable Guide Catheter, averaged 125 minutes. The mean fluoroscopy duration was 46 minutes. Fluoroscopy time was a relatively short proportion of the overall Procedure Time (29%). The MitraClip device was implanted successfully in a majority (95.3%) of patients. There were no intra-procedural deaths.

Table 5: Prohibitive Risk Primary MR MitraClip™ Cohort – Procedural Results

Procedural Result [a]	Mean±SD (N) Median (Min, Max)[a]
Procedure Time[b] (min)	157±81 (124) 134 (39, 524)
Device Time[c] (min)	125±75 (124) 110 (9, 511)
Fluoroscopy Duration (min)	46±26 (126) 39 (3, 167)
# of Devices Implanted 0 1 2	 4.7% (6/127) 44.1% (56/127) 51.2% (65/127)

[a] Sample sizes or denominators smaller than 127 reflect missing data.
[b] Procedure time is measured from the time the transseptal procedure starts until the time the Steerable Guide Catheter is removed.
[c] Device time is measured from the time the Steerable Guide Catheter is placed in the intra-atrial septum until the time the MitraClip™ Delivery System is retracted into the Steerable Guide Catheter.

At 12 months, MAEs occurred at a rate of 35.4% among Prohibitive Risk primary MR MitraClip™ patients, with deaths (23.6%) and transfusions (19.7%) comprising the majority of events (**Table 6**). The rate of stroke was 2.4% and rate of non-elective cardiovascular surgery was 0.8% at 12 months.

Table 6: Prohibitive Risk Primary MR MitraClip™ Cohort – CEC Adjudicated Major Adverse Events at 30 Days and 12 Months

Description of Event	Prohibitive Risk Primary MR MitraClip™ Patients (N = 127)	
	30 Days % (n/N)	12 Months % (n/N)
Death	6.3% (8/127)	23.6% (30/127)
Myocardial infarction	0.8% (1/127)	0.8% (1/127)
Re-operation for failed surgical repair or replacement	0	0
Non-elective cardiovascular surgery for adverse events	0.8% (1/127)	0.8% (1/127)
Stroke	2.4% (3/127)	2.4% (3/127)
Renal Failure	1.6% (2/127)	3.9% (5/127)
Deep wound infection	0	0.0% (0/127)
Ventilation > 48 hours	3.1% (4/127)	4.7% (6/127)
GI complication requiring surgery	0.8% (1/127)	2.4% (3/127)
New onset of permanent AF	0	0.0% (0/127)
Septicemia	0	4.7% (6/127)
Transfusion ≥ 2 units	12.6% (16/127)	19.7% (25/127)
Total[a]	**18.9% (24/127)**	**35.4% (45/127)**
Total[a] (Excluding Transfusions ≥ 2 units)	**9.4% (12/127)**	**26.0% (33/127)**

[a] Total number of patients may not equal the sum of patients in each row since one patient may experience multiple events.

Other secondary safety endpoints occurred at a relatively low rate (**Table 7**), consistent with access to the mitral valve achieved via the femoral vein and inferior vena cava. Major vascular complications occurred in 5.5% of patients at 30 days and in 7.1% of patients at 12 months. Major bleeding complications, defined as procedure-related bleeding requiring transfusions of at least 2 units or surgery, occurred at a rate of 12.6% at 30 days, with a majority of events requiring transfusions rather than surgery. Bleeding events that occurred after 30 days were unrelated to the MitraClip™ procedure. Clinically significant atrial septal defect requiring treatment occurred at a rate of 2.4% at 12 months. A low rate (2.4%) of mitral stenosis was observed at 12 months, with a total of 3 patients reported to have experienced mitral stenosis defined as Echocardiography Core Laboratory assessed mitral valve area less than 1.5 cm^2 through 12 months. The site did not report mitral stenosis for these patients and none of these patients underwent mitral valve surgery for stenosis.

Table 7: Prohibitive Risk Primary MR Cohort – Other Secondary Safety Events at 30 Days and 12 Months

Description of Event	30 Days % (n/N)	12 Months % (n/N)
Major Vascular Complications	5.5% (7/127)	7.1% (9/127)
Major Bleeding Complications	12.6% (16/127)	15.7% (20/127)
Non-Cerebral Thromboembolism	1.6% (2/127)	1.6% (2/127)
New Onset of Persistent Atrial Fibrillation	3.9% (5/127)	3.9% (5/127)
Heart Block / Other Arrhythmia requiring Permanent Pacemaker	0.0% (0/127)	1.6% (2/127)
Endocarditis	0.0% (0/127)	0.0% (0/127)
Thrombosis	0.0% (0/127)	0.0% (0/127)
Hemolysis	0.0% (0/127)	0.0% (0/127)
Atrial Septal Defect	1.6% (2/127)	2.4% (3/127)
Mitral Valve Stenosis	0.0% (0/127)	2.4% (3/127)

The Duke University Medical Center database, which consists of patient-level data with echocardiographic, medical history and follow-up data on a large number of patients with MR ≥ 3+ provides a descriptive comparator for mortality. This database allowed for characterization of survival in patients deemed high risk for surgery and managed non-surgically at the Duke University Medical Center despite clear Class I indications for surgery according to the ACC/AHA Guidelines for the Management of Patients with Valvular Heart Disease. Nine hundred and fifty-three (953) patients in the Duke database with 3+ or 4+ MR were identified as too high risk for surgery based on the same high risk criteria as those in the EVEREST II HRR and REALISM studies (i.e., STS mortality risk ≥ 12% or protocol-specified surgical risk factors) and managed non-surgically. This made up the Duke High Risk Cohort, of which 65 patients were identified as primary MR. **Table 8** shows both groups were comprised of elderly patients, with a majority of patients over the age of 75 years. The Duke High Risk primary MR Cohort reported a lower LVEF at baseline and a higher proportion of female patients than the Prohibitive Risk primary MR Cohort. The Prohibitive Risk primary MR Cohort reported a higher proportion of patients with COPD and NYHA III/IV symptoms at baseline. Both groups had high rates of previous MI, atrial fibrillation, and previous cardiovascular surgery.

Table 8: Baseline and Demographic Characteristics – Prohibitive Risk Primary MR and Duke High Risk Primary MR Cohorts

Baseline Characteristic	Prohibitive Risk Primary MR MitraClip Cohort % (n/N) (N = 127)	Duke High Risk Primary MR Medical Therapy Cohort % (n/N) (N = 65)
Age (years), Mean±SD (N)	82.4±8.7 (127)	76.8±11.3 (65)
Patients over 75 years of age	83.5% (106/127)	67.7% (44/65)
Male Gender	55.1% (70/127)	36.9% (24/65)
Body Mass Index (kg/m2), Mean±SD (N)	25.0±5.7 (127)	25.4±5.0(65)
Prior Myocardial Infarction	24.4% (31/127)	33.8% (22/65)
Atrial Fibrillation History	70.5% (86/122)	58.5% (38/65)
Prior Stroke	10.2% (13/127)	18.5% (12/65)
COPD with Home Oxygen	13.4% (17/127)	6.2% (4/65)
Hypertension	88.2% (112/127)	75.4% (49/65)
Diabetes	29.9% (38/127)	36.9% (24/65)
Moderate to Severe Renal Disease	28.3% (36/127)	20.0% (13/65)
Previous Cardiovascular Surgery	48.0% (61/127)	56.9% (37/65)
Previous Percutaneous Coronary Intervention	33.3% (42/126)	58.5% (38/65)
NYHA Functional Class III/IV	86.6% (110/127)	43.8% (28/65)
STS Predicted Mortality Risk	13.2±7.3 (127)	13.3±9.0
LV Ejection Fraction (%), Mean±SD (N)	60.6±9.5 (112)	44.9±11.7 (65)
LV Internal Diameter, systole (cm), Mean±SD (N)	3.4±0.8 (113)	3.4±0.9 (65)

Table 9 displays the Kaplan-Meier estimates of freedom from survival in the Prohibitive Risk primary MR patients to the Duke High Risk primary MR patients. Based on these Kaplan-Meier curves, mortality in the Prohibitive Risk primary MR Cohort was 6.4% at 30 days and 24.8% at 12 months compared to 10.9% at 30 days and 30.6% at 12 months in the Duke High Risk primary MR patients. While these results are descriptive and limited by differences described above, they suggest that there is no elevated risk of mortality in Prohibitive Risk primary MR patients who undergo the MitraClip™ procedure over non-surgical management.

Table 9: Number at Risk, Kaplan-Meier Estimates and 95% CIs

Time Post Index Procedure	Baseline	30 Days	6 Months	12 Months
Prohibitive Risk Primary MR MitraClip Patients (N = 127)				
# At Risk	127	117	106	85
# Events	0	8	19	31
% Event Free	100%	93.6%	84.8%	75.2%
95% CI[a]	-	[87.6%, 96.8%]	[77.2%, 90.0%]	[66.1%, 82.1%]
Duke High Risk Primary MR Medical Therapy Patients (N = 65)				
# At Risk	65	57	49	39
# Events	0	7	13	19
% Event Free	100%	89.1%	79.6%	69.4%
95% CI[a]	-	[78.5%, 94.7%]	[67.4%, 87.6%]	[56.3%, 79.3%]

[a] Confidence intervals are provided to illustrate the variability of the corresponding summary statistic. They should not be used to draw statistical inference.

MR severity at baseline, discharge and 12 months are presented in **Table 10** for patients with data available at each follow-up (Completers Analysis). Immediate improvement in MR severity was noted at discharge with 82.1% and 53.7% of surviving patients reporting MR severity ≤ 2+ and ≤ 1+, respectively. This improvement was sustained at 12 months, with the majority (83.3%) of surviving patients reporting MR severity ≤ 2+ and 36.9% reporting MR severity ≤ 1+. At 12 months, freedom from death and MR > 2+ was 61.4% and freedom from death and MR > 1+ was 27.2% patients.

Table 10: Prohibitive Risk Primary MR MitraClip™ Cohort – MR Severity at Baseline and Follow-up Completers Analysis

MR Severity	Baseline % (n/N)	Discharge[a] % (n/N)	12 Months % (n/N)
0 : None	0	1.6% (2/123)	0
1+: Mild	0	52.0% (64/123)	36.9% (31/84)
2+: Moderate	9.7% (12/124)	28.5% (35/123)	46.4% (39/84)
3+: Moderate-to-severe	58.9% (73/124)	13.0% (16/123)	13.1% (11/84)
4+: Severe	31.5% (39/124)	4.9% (6/123)	3.6% (3/84)
Missing	3	3	13
Death	0	1	30
MR ≤ 2+ in surviving patients	9.7% (12/124)	82.1% (101/123)	83.3% (70/84)
MR ≤ 1+ in surviving patients	0.0% (0/124)	53.7% (66/123)	36.9% (31/84)
Freedom from Death and MR > 2+	9.7% (12/124)	81.5% (101/124)	61.4% (70/114)
Freedom from Death and MR > 1+	0.0% (0/124)	53.2% (66/124)	27.2% (31/114)

[a] 30-day MR severity was used if discharge MR was unavailable.

Reduced preload as a result of MR reduction achieved with the MitraClip™ device resulted in reverse left ventricular remodeling (**Table 11**), characterized largely by a clinically important decrease in diastolic volume (-16.6 ml) and dimension (-0.2 cm).

Table 11: Prohibitive Risk Primary MR MitraClip™ Cohort – LV Measurements at Baseline and 12 Months Patients with Paired Data[a]

LV Measurement	N	Baseline	12-month	Difference (12-month – Baseline)	%Change (12-month – Baseline)
LVEDV, ml					
Mean±SD	69	125.1±40.1	108.5±37.9	-16.6±22.9	-11.5±17.9
Median		119.7	104.7	-12.3	-10.2
95% CI[b,c]				(-22.1, -11.1)	(-15.9, -7.2)
LVIDd, cm					
Mean±SD	80	5.0±0.6	4.8±0.6	-0.2±0.4	-3.7±8.2
Median		5.1	4.9	-0.2	-4.0
95% CI[b,c]				(-0.3, -0.1)	(-5.6, -1.9)
LVESV, ml					
Mean±SD	69	49.1±24.5	46.1±21.4	-3.0±13.7	-1.3±27.0
Median		45.7	41.0	-1.5	-2.7
95% CI[b,c]				(-6.3, 0.3)	(-7.7, 5.2)
LVIDs, cm					
Mean±SD	75	3.4±0.7	3.3±0.7	-0.1±0.5	-0.2±16.4
Median		3.2	3.3	-0.1	-2.3
95% CI[b,c]				(-0.2, 0.1)	(-4.0, 3.6)

[a] Only patients who had a measurement at both Baseline and 12 months are included.
[b] 95% CI is based on a t-distribution.
[c] Confidence intervals are provided to illustrate the variability of the corresponding summary statistic. They should not be used to draw statistical inference.

Improvement in LV function resulted in improvements in heart failure symptoms. NYHA Functional Class at baseline and follow-up are presented in **Table 12** for patients with data available at each follow-up (Completers Analysis). Immediate improvement in NYHA Class was noted at 30 days with 82.3% of surviving patients reporting NYHA Class I or II symptoms and sustained through 12 months (86.9% NYHA Class I or II). At 12 months, freedom from death and NYHA Class III or IV symptoms was 64.0%. This improvement in NYHA Class symptoms is clinically important given that the majority of these patients (86.6%) were enrolled with NYHA Class III or IV symptoms.

Table 12: Prohibitive Risk Primary MR MitraClip™ Cohort – NYHA Functional Class at Baseline and Follow-up Completers Analysis

NYHA Functional Class	Baseline % (n/N)	30 Days % (n/N)	12 Months % (n/N)
I	2.4% (3/127)	33.6% (38/113)	40.5% (34/84)
II	11.0% (14/127)	48.7% (55/113)	46.4% (39/84)
III	63.8% (81/127)	15.9% (18/113)	10.7% (9/84)
IV	22.8% (29/127)	1.8% (2/113)	2.4% (2/84)
Missing	0	5	13
Death	0	9	30
NYHA I/II in surviving patients	13.4% (17/127)	82.3% (93/113)	86.9% (73/84)
Freedom from Death and NYHA Class III/IV	13.4% (17/127)	76.2% (93/122)	64.0% (73/114)

Table 13 shows a mean change of +6.0 points in the Physical Component Summary (PCS) score and +5.6 points in the Mental Component Summary (MCS) score from baseline to 12 months after the MitraClip™ procedure. These changes are well above the 2-3 point minimally important difference (MID) threshold reported in the literature.

Table 13: Prohibitive Risk Primary MR MitraClip™ Cohort – SF-36 Quality of Life at Baseline and 12 Months Completers Analysis[a]

Component	N	Baseline	12-month	Difference (12-month – Baseline)
Physical Component Summary Score				
Mean±SD	73	33.4±8.6	39.4±10.5	6.0±8.6
Median		32.4	40.7	5.6
95% CI [b,c]				(4.0, 8.0)
Mental Component Summary Score				
Mean±SD	73	46.6±13.4	52.2±10.2	5.6±14.0
Median		49.8	54.0	3.2
95% CI [b,c]				(2.3, 8.9)

[a] Only patients who had a measurement at both Baseline and 12 months are included.
[b] 95% CI is based on a t-distribution.
[c] Confidence intervals are provided to illustrate the variability of the corresponding summary statistic. They should not be used to draw statistical inference.

A clinically important decrease in the rate of hospitalization for heart failure was observed following discharge from the MitraClip™ procedure (0.67 to 0.18 per patient-year, a 73% reduction, **Table 14**) between the pre-enrollment and the post-discharge 12-month periods.

Table 14: Prohibitive Risk Primary MR MitraClip™ Cohort – Heart Failure Hospitalizations

	12 months Pre-enrollment	Post-discharge through 12 months
# Patients for Analysis	127	120
# Patients with Events	48	13
# Events	85	17
Follow-up (Patient-Years)	127	97
Rate[a]	0.67	0.18
(95% Two-sided CI [a,b])	(0.54, 0.83)	(0.11, 0.28)
# days hospitalized (Mean±SD)	6.0±4.5	5.9±3.8

[a] CI is obtained from a Poisson regression model.
[b] Confidence intervals are provided to illustrate the variability of the corresponding summary statistic. They should not be used to draw statistical inference.

Effectiveness results demonstrate that 82.1% (101/123) of completers experienced MR reduction from 3+ or 4+ to 2+ or less at discharge following the MitraClip™ procedure. Reduction of MR at 12 months was sustained to ≤2+ in 83.3% (70/84), and to ≤1+ in 36.9% (31/84) of patients for whom echocardiographic data was available. Reduction in MR severity was associated with reverse left ventricular remodeling characterized largely by clinically important decreases in diastolic volume and dimension.

Patients also experienced clinically important improvement in NYHA Functional Class at 12 months; more than 80% of patients experienced NYHA Class III or Class IV symptoms at baseline, which reduced to less than 15% at 12 months. Despite the elderly and highly co-morbid nature of the population, quality of life as measured by the SF-36 quality of life physical and mental component scores showed clinically important improvement. Sensitivity analyses showed that these effectiveness results are robust to missing data. Finally, heart failure hospitalizations showed clinically important reduction in the 12 months post-MitraClip™ procedure from the 12 months pre-MitraClip™ procedure, including in a sensitivity analysis where death is included in the analysis as a heart failure hospitalization.

Reduction in MR severity was assessed in patients who have 2-year follow-up available. **Table 15** shows that MR reduction in surviving patients to ≤ 2+ and ≤ 1+ is 82.5% (33/40) and 35.0%, (14/40) respectively, at 2 years. Therefore, there is no evidence of deterioration of MR severity from 12 months to 2 years in surviving patients.

Table 15: Prohibitive Risk Primary MR MitraClip™ Cohort – Durability of MR Reduction

MR Severity	Baseline % (n/N)	12 Months % (n/N)	2 Years % (n/N)
0 : None	0	0	0
1+: Mild	0	36.9% (31/84)	35.0% (14/40)
2+: Moderate	9.7% (12/124)	46.4% (39/84)	47.5% (19/40)
3+: Moderate-to-severe	58.9% (73/124)	13.1% (11/84)	15.0% (6/40)
4+: Severe	31.5% (39/124)	3.6% (3/84)	2.5% (1/40)
MR ≤ 2+ in surviving patients	9.7% (12/124)	83.3% (70/84)	82.5% (33/40)
MR ≤ 1+ in surviving patients	0.0% (0/124)	36.9% (31/84)	35.0% (14/40)

Based on the outcomes of Prohibitive Risk Primary MR patients treated with the MitraClip System, the totality of evidence demonstrates reasonable assurance of safety and effectiveness of MitraClip™ to reduce MR and provide patient benefit in this discreet and specific patient population.

29.2.2 COAPT Clinical Results

The COAPT Trial was a prospective, randomized (1:1; MitraClip™ + GDMT vs. GDMT alone), open-label, multicenter investigational study intended to demonstrate: (1) MitraClip™ was safe in subjects with secondary MR, and (2) MitraClip™ could reduce recurrent HF hospitalization as compared to the GDMT Control group. Additional details of the study design can be found in **Section 29.1** and **Table 2**. This study is registered at clinicaltrials.gov (NCT01626079). Results from this study have been previously published.[5], [6] Results described below reflect data collected through September 21, 2022.

29.2.2.1 COAPT Accountability of PMA Cohort

At the time of database lock, a total of 614 subjects were randomized in this trial, including 302 Device subjects and 312 Control subjects.

There were four different analysis populations defined in the protocol: Intention-to-Treat (ITT) population, Per Protocol (PP) population, As Treated (AT) population, and Safety Analysis (SA) population, as summarized in **Table 16** and **Figure 2**. The primary analysis for safety was the Safety Analysis, and that for effectiveness was the ITT analysis.

Table 16: Analysis Populations

Analysis Population	Definition	Number of Patients	
		Device	Control
Intention-to-Treat (ITT)	All randomized subjects	302	312
As Treated (AT)	Randomized subjects who received the treatment as randomized	294	320
Per Protocol (PP)	Subjects who met major inclusion and none of the major exclusion criteria and received the treatment as randomized	270	289
Safety Analysis (SA)	All ITT subjects in the Device group with an attempted implant procedure*	293	

*Attempted implant procedure is defined as administration of anesthesia for the MitraClip procedure.

29.2.2.2 COAPT Study Population Demographics and Baseline Parameters

The demographics and baseline characteristics of the study population are typical for an HF study performed in the U.S., as shown in **Table 17**. The two study groups were well-balanced, with no significant difference in patient demographics and baseline characteristics.

Figure 2: Disposition of COAPT Randomized Subjects

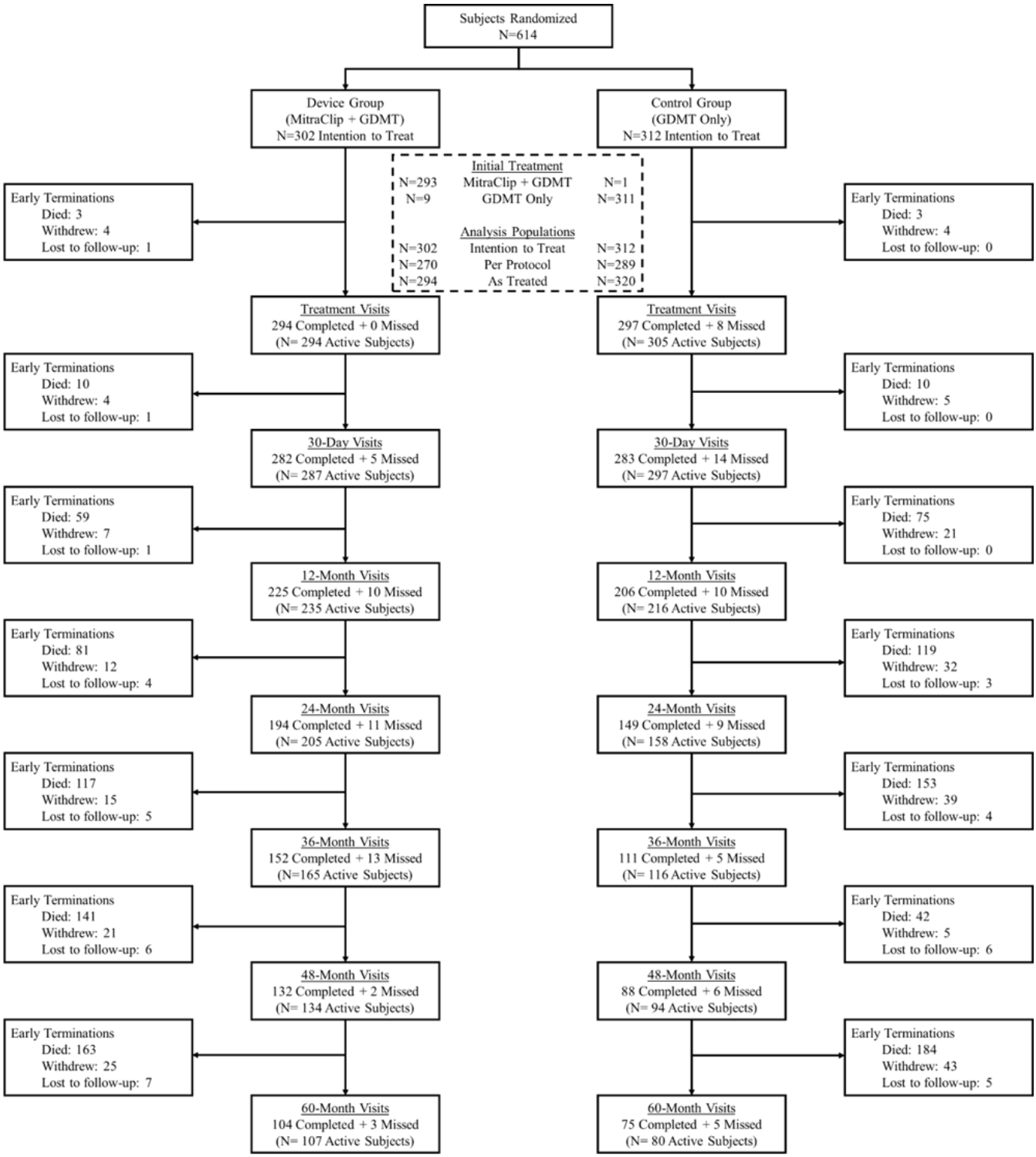


Table 17: Patient Demographics and Baseline Characteristics (ITT Population)

Demographics and Baseline Characteristics	Summary Statistics*		P-Value†
	Device (N=302)	Control (N=312)	
Age at Registration (year)	71.7 ± 11.8 (302)	72.8 ± 10.5 (312)	0.22
Male	66.6% (201/302)	61.5% (192/312)	0.20
Race/Ethnicity			
White or Caucasian	74.5% (225/302)	74.4% (232/312)	0.97
Non-white	25.5% (77/302)	25.6% (80/312)	
Height (cm)	170.8 ± 10.4 (301)	169.9 ± 10.8 (306)	0.25
Weight (kg)	78.8 ± 17.2 (301)	78.4 ± 20.1 (307)	0.80
Body Mass Index (kg/m^2)	27.0 ± 5.8 (300)	27.1 ± 5.9 (305)	0.99
Serum Creatinine (mg/dL)	1.8 ± 1.2 (300)	1.8 ± 1.4 (306)	0.84
Creatinine Clearance (mL/min)	50.9 ± 28.5 (299)	47.8 ± 25.0 (302)	0.16
Creatinine Clearance ≤ 60 mL/min	71.6% (214/299)	75.2% (227/302)	0.32
BNP (pg/mL)	1014.8 ± 1086.0 (208)	1017.1 ± 1212.8 (209)	0.98
NT-proBNP (pg/mL)	5174.3 ± 6566.6 (74)	5943.9 ± 8437.6 (85)	0.52
Elevated BNP or NT-proBNP prior to Enrollment	93.4% (267/286)	93.1% (282/303)	0.89
Extremely High Risk for MV Surgery	68.6% (205/299)	69.9% (218/312)	0.73
KCCQ Overall Summary Score	53.2 ± 22.8 (302)	51.6 ± 23.3 (309)	0.39
Six Minute Walk Test Distance (meters)	249.6 ± 123.8 (296)	234.5 ± 123.5 (305)	0.13
NYHA Functional Class			
Class I	0.3% (1/302)	0.0% (0/311)	0.49
Class II	42.7% (129/302)	35.4% (110/311)	0.06
Class III	51.0% (154/302)	54.0% (168/311)	0.45
Class IV	6.0% (18/302)	10.6% (33/311)	0.04
SF-36 Quality of Life Physical Component Score	33.0 ± 9.1 (299)	32.6 ± 10.0 (308)	0.63
SF-36 Quality of Life Mental Component Score	46.7 ± 12.7 (299)	45.3 ± 13.0 (308)	0.19
Cardiovascular Event History			
Ischemic Cardiomyopathy	60.9% (184/302)	60.6% (189/312)	0.93
Non-Ischemic Cardiomyopathy	39.1% (118/302)	39.4% (123/312)	
Prior TIA	8.6% (26/302)	5.4% (17/312)	0.13
Prior Stroke	12.3% (37/302)	11.2% (35/312)	0.69
Prior Stroke or TIA	18.5% (56/302)	15.7% (49/312)	0.35
Prior Myocardial Infarction	51.7% (156/302)	51.3% (160/312)	0.93
Coronary Artery Disease (CAD)	72.2% (218/302)	73.1% (228/312)	0.80
Hypertension	80.5% (243/302)	80.4% (251/312)	0.99
Hypercholesterolemia	55.0% (166/302)	52.2% (163/312)	0.50

Demographics and Baseline Characteristics	Summary Statistics*		P-Value†
	Device (N=302)	Control (N=312)	
Angina	16.9% (51/302)	23.4% (73/312)	0.04
Chronic Obstructive Pulmonary Disease	23.5% (71/302)	23.1% (72/312)	0.90
Arrhythmia Event History	66.6% (201/302)	64.4% (201/312)	0.58
Ventricular Fibrillation	5.6% (17/302)	8.0% (25/312)	0.24
Ventricular Flutter	0.0% (0/302)	0.0% (0/312)	0.99
Ventricular Tachycardia	24.8% (75/302)	22.4% (70/312)	0.48
Atrial Flutter	10.3% (31/302)	10.9% (34/312)	0.80
Atrial Fibrillation	55.6% (168/302)	51.0% (159/312)	0.25
Atrial Fibrillation or Flutter	57.3% (173/302)	53.2% (166/312)	0.31
Any Hospitalization 12 months prior to enrollment	67.5% (204/302)	65.1% (203/312)	0.51
Heart Failure	58.3% (176/302)	56.1% (175/312)	0.58
Other-Cardiovascular	11.6% (35/302)	9.3% (29/312)	0.35
Non-Cardiovascular	7.9% (24/302)	7.1% (22/312)	0.67
Co-morbidity			
Diabetes	35.1% (106/302)	39.4% (123/312)	0.27
Peripheral Vascular Disease	17.2% (52/302)	18.3% (57/312)	0.73
Renal Disease	57.0% (172/302)	56.7% (177/312)	0.96
History of Anemia	22.5% (68/302)	24.4% (76/312)	0.59
History of Major Bleeds or Bleeding Disorder	7.6% (23/302)	7.1% (22/312)	0.79
STS Replacement Score (%)	7.8 ± 5.5 (302)	8.5 ± 6.2 (312)	0.16
STS Repair Score (%)	5.6 ± 5.6 (302)	6.0 ± 5.4 (312)	0.39
Prior Cardiac Interventions			
Coronary Artery Bypass Craft (CABG)	40.1% (121/302)	40.4% (126/312)	0.94
PTCA/Stents/Atherectomy	43.0% (130/302)	49.0% (153/312)	0.14
Device Implantation			
None	33.1% (100/302)	33.0% (103/312)	0.98
ICD	30.1% (91/302)	32.4% (101/312)	0.55
CRT-P	1.7% (5/302)	1.9% (6/312)	0.80
CRT-D	36.4% (110/302)	33.0% (103/312)	0.37
Pacemaker	6.0% (18/302)	8.0% (25/312)	0.32
Defibrillator (ICD or CRT-D)	62.6% (189/302)	61.5% (192/312)	0.79
Resynchronization (CRT-D or CRT-P)	38.1% (115/302)	34.9% (109/312)	0.42
Pacing (CRT-P or Pacemaker)	7.3% (22/302)	9.9% (31/312)	0.24
Prior Cardiac Valve Interventions			
Aortic Valve Intervention	3.3% (10/302)	4.5% (14/312)	0.45
Pulmonic Valve Intervention	0.0% (0/302)	0.0% (0/312)	0.99
Tricuspid Valve Intervention	0.0% (0/302)	0.0% (0/312)	0.99
Mitral Valve Intervention	0.3% (1/302)	0.0% (0/312)	0.49

Demographics and Baseline Characteristics	Summary Statistics*		P-Value†
	Device (N=302)	Control (N=312)	
Echocardiographic Core Laboratory Measures			
Mitral regurgitation severity			
3+: Moderate-to-Severe	49.0% (148/302)	55.3% (172/311)	0.12
4+: Severe	51.0% (154/302)	44.7% (139/311)	
Effective Regurgitant Orifice Area (EROA, cm^2)	0.41 ± 0.15 (289)	0.40 ± 0.15 (302)	0.42
Left Ventricular Ejection Fraction (LVEF, %)	31.3 ± 9.1 (281)	31.3 ± 9.6 (294)	0.97
≤ 40 %	82.2% (231/281)	82.0% (241/294)	0.94
Left Ventricular End Systolic Dimension (LVESD, cm)	5.3 ± 0.9 (301)	5.3 ± 0.9 (306)	0.82
Left Ventricular End Diastolic Dimension (LVEDD, cm)	6.2 ± 0.7 (301)	6.2 ± 0.8 (307)	0.80
Left Ventricular End Systolic Volume (LVESV, mL)	135.5 ± 56.1 (281)	134.3 ± 60.3 (294)	0.81
Left Ventricular End Diastolic Volume (LVEDV, mL)	194.4 ± 69.2 (281)	191.0 ± 72.9 (294)	0.57
LVEDV Index (mL/m^2)	102.3 ± 33.7 (279)	100.6 ± 35.0 (288)	0.56
Right Ventricular Systolic Pressure (RVSP, mmHg)	44.0 ± 13.4 (253)	44.6 ± 14.0 (275)	0.61
Medication Use at Baseline			
Beta-blocker	91.1% (275/302)	89.7% (280/312)	0.58
ACEI, ARB or ARNI	71.5% (216/302)	62.8% (196/312)	0.02
Mineralocorticoid receptor antagonist	50.7% (153/302)	49.7% (155/312)	0.81
Nitrate	6.3% (19/302)	8.0% (25/312)	0.41
Hydralazine	16.6% (50/302)	17.6% (55/312)	0.72
Diuretic	89.4% (270/302)	88.8% (277/312)	0.80
Chronic oral anticoagulant	46.4% (140/302)	40.1% (125/312)	0.12
Aspirin	57.6% (174/302)	64.7% (202/312)	0.07
P2Y12 receptor inhibitor	25.2% (76/302)	22.8% (71/312)	0.48
Statin	62.6% (189/302)	60.6% (189/312)	0.61

*Continuous measures – Mean ± SD; categorical measures - % (no./total no.).
†P-values are from *t*-test for continuous variables and from Chi-square test or Fisher's exact test when Cochran's rule is not met for categorical variables. All p-values displayed are two-sided and for information only.

29.2.2.3 COAPT Safety and Effectiveness Results

Primary Safety Endpoint

The rate of freedom from device-related complications at 12 months was 96.6%, with a lower 95% confidence limit of 94.8%, which was higher than the pre-specified performance goal of 88% (p<0.0001). As such, the COAPT Trial met its primary safety endpoint. A breakdown of the composite primary safety endpoint events is presented in **Table 18**.

Table 18: Outcomes of the Primary Safety Endpoint Components (SA Population)

Event	Summary Statistics* (N = 293)
Device-related complications at 12 months	9 (3.4%)
-- Single leaflet device attachment	2 (0.7%)
-- Device embolization	1 (0.3%)
-- Endocarditis requiring surgery	0 (0.0%)
-- Mitral stenosis requiring surgery	0 (0.0%)
-- LVAD implant	3 (1.2%)
-- Heart transplant	2 (0.8%)
-- Any device-related complication requiring non-elective cardiovascular surgery	1 (0.3%)

*# events (Kaplan-Meier rate)

Primary Effectiveness Endpoint

A total of 160 and 283 HF hospitalizations occurred within 24 months in the Device and Control groups, respectively. The annualized rates (events per patient-year) of HF hospitalization were 0.358 in the Device group and 0.679 in the Control group, with a hazard ratio (HR) of 0.525 (upper 95% confidence limit: 0.664), representing a 47.5% reduction in the risk of recurrent HF hospitalization by the Joint Frailty Model in favor of the Device (p<0.0001), as summarized in **Table 19**. Therefore, the COAPT Trial met its primary effectiveness endpoint. The successes of the primary safety endpoint and the primary effectiveness endpoint were confirmed by the AT analysis, PP analysis, and sensitivity analysis.

Table 19: Recurrent HF Hospitalization through 24 Months – Primary Effectiveness Endpoint (ITT Population)

	Device (N=302)	Control (N=312)
Number of Subjects*	92 (30.5%)	151 (48.4%)
Number of Events	160	283
Total Follow-Up (patient-years)*	446.5	416.8
Annualized Rate	0.358	0.679
[95% CI]†	[0.307, 0.418]	[0.604, 0.763]
Joint Frailty Model		
Hazard Ratio – Device vs. Control [95% CI]	0.525 [-, 0.664]	
Relative Risk Reduction – Device vs. Control [95% CI]	0.475 [0.336, -]	
P-Value	< 0.0001	

Powered Secondary Endpoints

Hypothesis testing was performed on 10 pre-specified secondary endpoints using a hierarchical test procedure, as shown in **Table 20**. All powered secondary endpoints were met.

Table 20: Summary of Hierarchical Secondary Endpoints (ITT Population)

Secondary Endpoints (hierarchical order)	Device (N=302)	Control (N=312)	HR or Difference [95% CI]	Lower 95% confidence limit	P-Value*
#1 Proportion of MR Severity ≤ 2+ at 12 Months; % (no./total no.) [95% CI]	94.8% (199/210) [90.82%, 97.36%]	46.9% (82/175) [39.29%, 54.53%]	-	-	< 0.0001
#2 All-Cause Mortality at 12 Months (Non-inferiority);† Kaplan-Meier estimate (SE) of event rate	19.1% (2.3%)	23.2% (2.4%)	0.809 [-, 1.085]	-	0.0003
#3 Finkelstein-Schoenfeld Analysis of a Hierarchical Composite of All-Cause Mortality and Recurrent HF Hospitalization through 24 Months	-	-	-	-	< 0.0001
#4 Change in KCCQ Overall Summary Score at 12 Months over Baseline; least square means (SE) [95% CI]	12.50 (1.82) [8.93, 16.08]	-3.56 (1.85) [-7.21, 0.08]	16.07 [10.97, 21.17]	-	< 0.0001
#5 Change in 6MWD at 12 Months over Baseline; least square means (SE) [95% CI]	-2.17 (9.12) [-20.10, 15.76]	-60.03 (8.99) [-77.69, -42.36]	57.86 [32.67, 83.05]	-	< 0.0001
#6 All-Cause Recurrent Hospitalizations through 24 Months;† annualized rate [95% CI]	1.062 [0.970, 1.162]	1.464 [1.352, 1.585]	0.760 [0.602, 0.960]	-	0.0213
#7 Proportion of NYHA Functional Class of I/II at 12-Month; % (no./total no.) [95% CI]	72.2% (171/237) [65.98%, 77.76%]	49.6% (115/232) [42.96%, 56.19%]	-	-	< 0.0001
#8 Change in Left Ventricular End Diastolic Volume at 12 Months over Baseline; least square means (SE) [95% CI]	-3.71 (5.08) [-13.71, 6.28]	17.06 (5.10) [7.03, 27.08]	-20.77 [-34.93, -6.62]	-	0.0041
#9 All-Cause Mortality through 24 Months;† Kaplan-Meier estimate (SE) of event rate	29.1% (2.8%)	46.1% (3.2%)	0.615 [0.463, 0.816]	-	0.0008
#10 Estimate of Freedom from All-Cause Mortality, Stroke, MI or Non-Elective Cardiovascular Surgery for Device-Related Complications at 30 Days; % (no./total no.)	96.9% (284/293)	-	-	94.7%	<0.0001

*All p-values were tests for superiority, except for the secondary endpoint of mortality at 12 months (#2), which was a test for non-inferiority, and for the secondary endpoint of freedom from composite of all-cause mortality, stroke, MI or non-elective cardiovascular surgery for device-related complications at 30 days (#10), which was compared against a performance goal.
†Analyzed when the last subject completed the 12-month follow-up.
Note: (1) Imputation of worst clinical outcomes for subjects experiencing HF death prior to 12 months for the changes in KCCQ, 6MWD, LVEDV and NYHA class. (2) Continuous endpoints (KCCQ, 6MWD, and LVEDV) were analyzed using Analysis of Covariance (ANCOVA). (3) HR – Hazard Ratio; CI – Confidence Interval; SE – Standard Error.

Long-term Endpoints up to 60 Months:

The 60-month COAPT results were evaluated on an As-Treated (AT) population. Subjects in the Control group were allowed to cross over to the Device group after primary endpoint analysis at 24-months. The crossover patients are included in the control group for this analysis. A summary of long-term outcomes through 60 months are detailed in **Table 21**.

The rate of freedom from device-related complications through 60 months for subjects in the Device group was 89.2%. All occurrences of SLDA, device embolization and device-related complications requiring non-elective CV surgery were within the first 30 days.

At 60-months, the Kaplan-Meier estimate of all-cause mortality event rate was 57.3% (freedom from event rate of 42.7%) for the Device group, and 67.2% (freedom from event rate of 32.8%) for the Control group. At 60-months, the Kaplan-Meier estimate of first heart failure hospitalization event rate was 61.2% (freedom from event rate of 38.8%) for the Device group, compared to 82.4% (freedom from event rate of 17.6%) for the Control group

At 60 months, 94.7% of subjects in the Device Group achieved ≤ 2+ MR compared to 91.3% of subjects in the Control Group, as shown in **Figure 3**. At 60 months, subjects in the Device Group were more commonly in NYHA Class I or II compared to those in the Control Group (24.1% vs 15.7%), as shown in **Figure 4**.

Table 21: Summary of Long-term Endpoints through 60 months

Descriptive Long-Term Endpoints up to 60 Months	Summary of Outcomes	
	Device	Control
Safety Composite: Device-related complications at 60 months*	23 (10.8%)	NA
-- Single leaflet device attachment	2 (0.7%)	NA
-- Device embolization	1 (0.3%)	NA
-- Endocarditis requiring surgery	0 (0.0%)	NA
-- Mitral stenosis requiring surgery	0 (0.0%)	NA
-- LVAD implant	13 (6.5%)	NA
-- Heart transplant	9 (4.7%)	NA
-- Any device-related complication requiring non-elective cardiovascular surgery	1 (0.3%)	NA
All-cause mortality through 60 months*	162 (57.3%)	189 (67.2%)
HF related hospitalization through 60 months*	151 (61.2%)	208 (82.4%)
Proportion of MR Severity Grade < 2+ at 60 months	94.7%	91.3%
NYHA Functional Class at 60 months	24.1%	15.7%

*# of Events (Kaplan-Meier Estimate)
NA = Not Applicable

Figure 3: Mitral Regurgitation Severity Through 60 Months

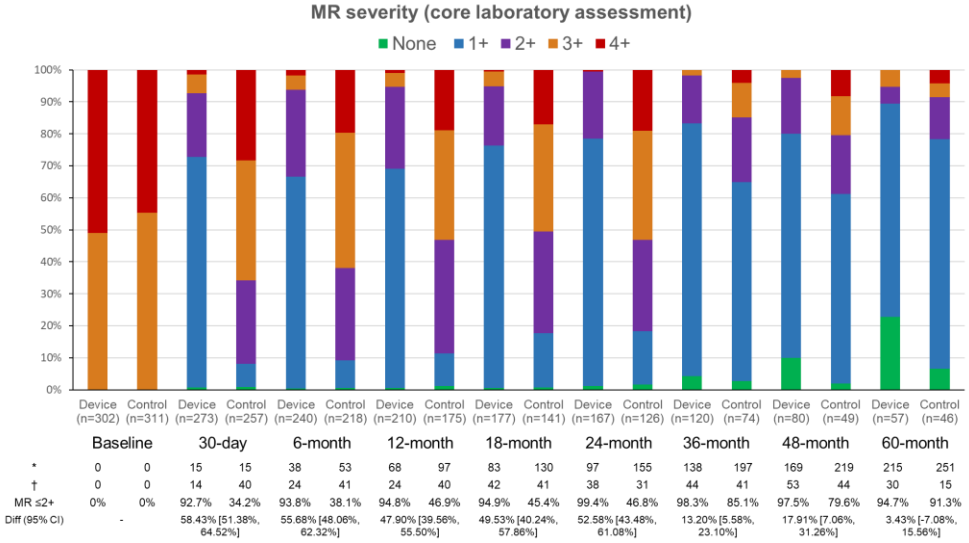
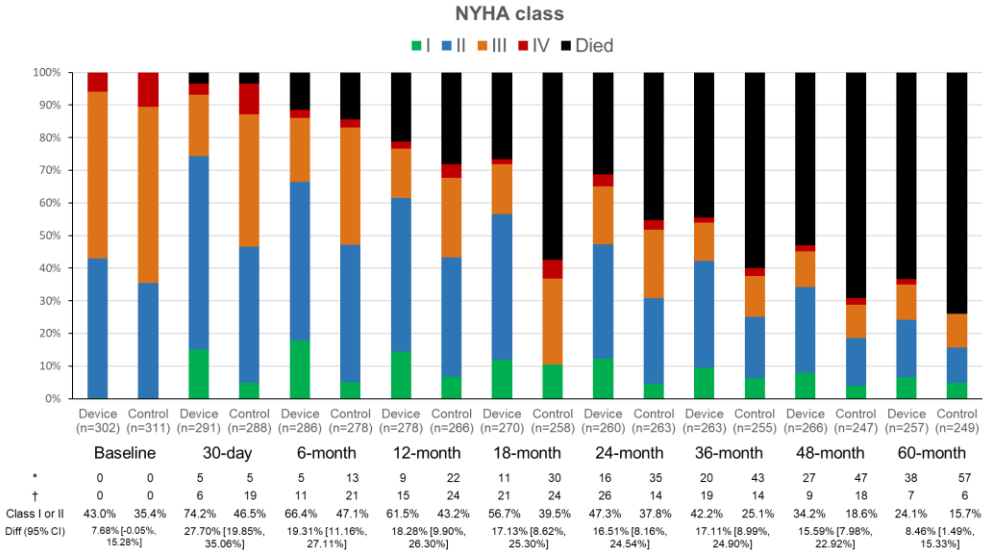


Figure 4: New York Heart Association Functional Class Through 60 Months



29.2.2.4 COAPT Adverse Events

The adverse events that occurred in the trial through 24 months are presented in **Table 22.**

Table 22: CEC-Adjudicated Adverse Events through 24 Months (SA Population)

Events	0-30 Days		0-12 Months		0-24 Months	
	Device	Control	Device	Control	Device	Control
All-cause mortality*	2.3% (7)	1.0% (3)	19.1% (57)	23.2% (70)	29.1% (80)	46.1% (121)
Cardiovascular	2.3% (7)	0.6% (2)	13.8% (40)	19.4% (57)	23.2% (60)	37.0% (93)
Heart failure	0.7% (2)	0.6% (2)	6.2% (17)	13.8% (39)	12.0% (28)	25.9% (61)
Stroke	0.7% (2)	0.0% (0)	2.9% (8)	2.9% (8)	4.4% (11)	5.1% (11)
Transient ischemic attack	0.0% (0)	0.0% (0)	1.1% (3)	1.1% (3)	1.1% (3)	1.1% (3)
Endocarditis requiring surgery	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)
ECL confirmed mitral stenosis requiring surgery	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)
LVAD implant	0.0% (0)	1.0% (3)	0.8% (2)	3.9% (11)	3.0% (6)	7.1% (16)
Heart transplant	0.0% (0)	0.0% (0)	0.8% (2)	2.2% (6)	1.4% (3)	3.6% (8)
Myocardial Infarction†	0.3% (1)	0.0% (0)	NA	NA	NA	NA
Major bleeding†	5.0% (15)	1.0% (3)	NA	NA	NA	NA
Iatrogenic ASD requiring intervention	0.7% (2)	NA	0.7% (2)	NA	0.7% (2)	NA
Device-related complications requiring non-elective CV surgery	0.3% (1)	NA	0.3% (1)	NA	0.3% (1)	NA

*Include adjudicated death events and deaths from the national death registry (for subjects who were lost to follow-up or withdrew from the COAPT study).
†Events were adjudicated up to 30 days post treatment visit.
Note: (1) Kaplan-Meier rate (# patients with events). Include only each patient's first occurrence of each event. (2) The follow-up duration was calculated from the randomization date. (3) ECL: Echocardiography Core Laboratory; LVAD: Left Ventricular Assists Device; ASD: Atrial Septal Defect; CV: Cardiovascular.
NA = Not Applicable or not adjudicated by CEC

29.2.2.5 COAPT Subgroup Analyses

Pre-specified Analyses:

The primary safety and primary effectiveness endpoints were examined across the following 4 subgroups:

- Sex (male vs. female)
- Etiology of cardiomyopathy (ischemic vs. non-ischemic)
- LVEF (> 40% vs. ≤ 40%)
- Extreme surgical risk status (yes vs. no, as determined by the Central Eligibility Committee)

There was no clinically significant difference among the subgroups for the primary safety outcome, and there were no clinically significant interaction effects between treatment and subgroups for the primary effectiveness outcome.

Post hoc Analyses:

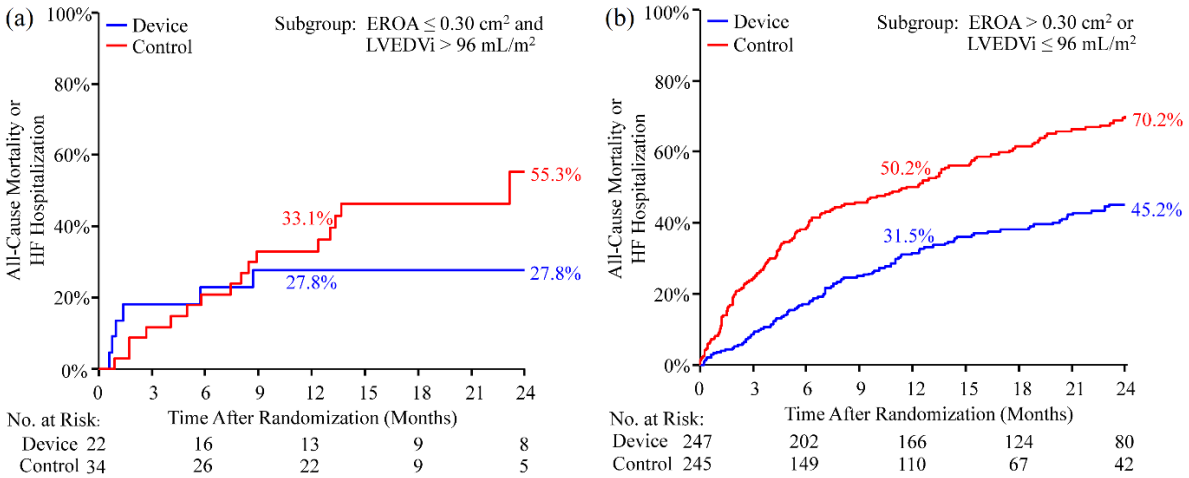
In light of publication of another study of the MitraClip device in literature, entitled "MITRA-FR Trial"[7], a comparison of the baseline characteristics of the subjects enrolled in the COAPT Trial and MITRA-FR Trial was performed to further define the proper patient population for the MitraClip SMR indication. The comparison suggested there were some differences between the two trials as shown in **Table 23**.

Table 23: Comparison in EROA and LVEDVi between Mitra-FR and COAPT

Baseline Characteristics	Mitra-FR	COAPT
EROA (mean ± SD; cm^2)	0.31 ± 0.11	0.41 ± 0.15
LVEDVi (mean ±SD; mL/m^2)	135 ± 35	101 ± 34

To explore whether there was any correlation between the clinical outcomes and the baseline EROA and LVEDVi, a *post hoc* subgroup analysis was conducted on the COAPT dataset, by comparing the composite rate of all-cause mortality or HF hospitalization between subjects with an EROA ≤ 0.30 cm^2 and an LVEDVi > 96 mL/m^2 and those with an EROA > 0.30 cm^2 or an LVEDVi ≤ 96 mL/m^2, where 0.30 was the lower bound of the EROA defining, along with other parameters, Grade III (or 3+) MR as per the 2017 ASE Recommendation for Noninvasive Evaluation of Native Valvular Regurgitation and 96 was the median LVEDVi value in the COAPT Trial.[8] A total of 22 subjects in the Device group and 34 subjects in the Control group had an EROA ≤ 0.30 cm^2 and an LVEDVi > 96 mL/m^2. The results of the subgroups analysis are shown in **Figure 5**. COAPT subjects with relatively less severe MR and larger left ventricles (**Figure 5a**) did not show a clinically meaningful benefit of the Device for all-cause mortality or HF hospitalization at the 12-month timepoint. For the remaining COAPT subjects (those with an EROA > 0.3cm^2 or an LVEDVI ≤ 96mL/m^2; **Figure 5b**), the difference in all-cause mortality or HF hospitalization seen in the overall population was maintained.

Figure 5: Subgroup Analysis Stratified by EROA and LVEDVi



Despite the absence of benefit of reduced all-cause mortality or HF hospitalization in the subgroup with an EROA ≤ 0.30 cm^2 and an LVEDVi > 96 mL/m^2, clinically meaningful improvements in the overall 6MWD (as shown in **Figure 6**; 11 subjects in the Device group and 26 subjects in the Control group had 6MWD values) and KCCQ (as shown in **Figure 7**; 15 subjects in the Device group and 27 subjects in the Control group had KCCQ values) compared to baseline were observed in Device group patients, an effect not observed in the same sub-population of the Control group. However, because of the nature of the *post hoc* subgroup analysis and the small sample size in the subgroup with an EROA ≤ 0.30 cm^2 and an LVEDVi > 96 mL/m^2, no statistical or clinical intra-group inferences can be made.

Figure 6: 6MWD for Subjects with an EROA ≤ 0.30 cm^2 and an LVEDVi > 96 mL/m^2

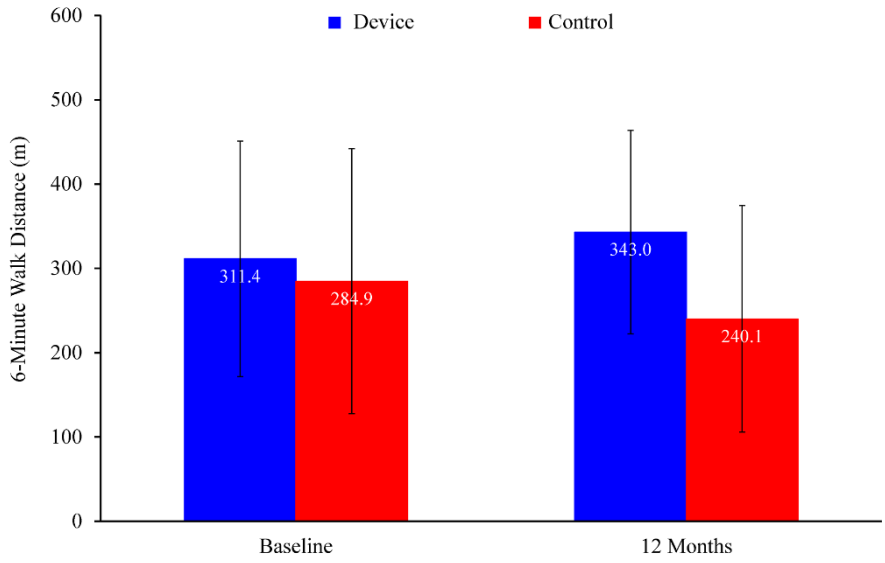
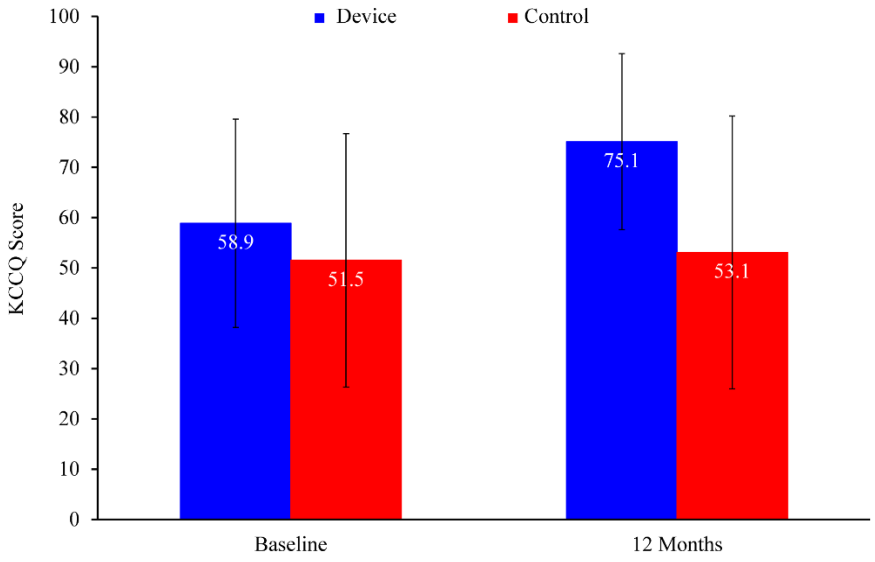


Figure 7: KCCQ Score for Subjects with an EROA ≤ 0.30 cm2 and an LVEDVi > 96 mL/m2



29.2.3 MitraClip Analysis Cohort (MAC) Clinical Results

The MitraClip™ Analysis Cohort (MAC) is a subset of patients treated with MitraClip entered into the TVT Registry who fulfill the following criteria:

- Significant symptomatic MR (≥ 3+) due to primary abnormality of the mitral apparatus, otherwise known as Primary (or degenerative) Mitral Regurgitation.
- Prohibitive risk for mitral valve surgery (MV) as determined by a heart team, which includes a cardiologist experienced in MV disease and a cardiac surgeon experienced in MV surgery, and existing comorbidities that would not preclude the expected benefit from reduction of the MR.
- Perform the 6MWT at baseline or unable to perform the 6MWT due to cardiac reasons.

29.2.3.1 MAC Subject Population

A total of 554 patients met eligibility criteria for this study between November 04, 2013, and July 29, 2015. The clinical follow up rate in the MAC study was 94.8% at 30 days and 78.2% at 12 months.

The patient demographic in the MAC dataset are as follows: mean age is 80.6 ± 9.8 years, 55.2% male/44.8 % female, 83.5% were NYHA Functional Class III/IV, mean STS valve replacement score of 11.17 ± 6.85%, and mean STS valve repair score of 8.74 ± 6.74 %. All patients had ≥3+ MR and 90.4% patients had only Primary MR etiology. The most common co-morbidities were heart failure within 2 weeks (79.7%), heart failure hospitalization in the past year (57.4%), hypertension (84.8%), and atrial fibrillation/flutter (66.7%).

29.2.3.2 MAC Safety and Effectiveness Results

Final Safety Findings (Key Endpoints)

The primary safety endpoint of MAC was met. The freedom from composite of death and device-related complications at 30-days post-procedure rate was 95.5% (529/554), with the lower one-sided 95% confidence limit of 93.8%, which was significantly greater than the pre-specified performance goal of 80% ($p < 0.0001$, one-sided exact binomial test).

The secondary safety endpoints of MAC were met.

- The estimated freedom from all-cause mortality at 12-months was 80.2% ± 1.85%, with the lower limit of one-sided 95% confidence interval of 77.2%, which is greater than the pre-specified performance goal of 66% (P<0.0001).
- Freedom from device-related complications at 12 months was 98.9% ± 0.45%, with the lower limit of one-sided 95% confidence interval of 98.2%, which is greater than the pre-specified performance goal of 90% (p < 0.0001).

Final Effectiveness Findings (Key Endpoints)

The primary effectiveness endpoint of MAC was also met. The 6MWT distance increased significantly at 12 months over baseline with an average of 31.6 ± 115.2m (n=184, one-sided p =0.0001, lower 95% CI 17.5m).

The secondary effectiveness endpoints of MAC were met.

- MR Severity at 12 months - Most (86.7%) of the patients went from ≥3+ MR at baseline to ≤2+ MR at 12-month follow-up (p<0.0001).
- Change in Left Ventricular End Diastolic Volume (LVEDV) at 12 months Over Baseline – LVEDV showed a mean reduction of 10.9 ± 48.9 ml (p=0.0111).

- Change in Left Ventricular Internal Diastolic Dimension (LVIDd) at 12 months Over Baseline – LVIDd showed a mean reduction of 0.22 ± 0.72 cm (p<0.0001).
- Change in KCCQ at 12 months over baseline – The KCCQ overall summary score showed a mean improvement of 30.4 ± 25.8 points (p<0.0001).
- NYHA Functional Class (FC) Improvement at 12 months – At 12 months post-procedure, 82.6% were NYHA FC I/II, as compared to 20.5% NYHA FC I/II at baseline (p<0.0001).

29.2.4 EXPAND Clinical Results

The MitraClip™ EXPAND study was a Prospective, Multi-Center, Single-Arm, International, Post Market, Observational Study designed to collect real-world data on the use of the next generation MitraClip™ NTR and MitraClip™ XTR Systems. Results presented here reflect data collected through the final data cutoff of August 5, 2020. Additional results have been published [9].

29.2.4.1 EXPAND Subject Population

A total of 1045 subjects were enrolled at 58 sites between April 26, 2018 and March 29, 2019. Enrollment in the study was closed on March 29, 2019. Of the 1045 subjects enrolled, 4 subjects did not have any procedure attempted.

29.2.4.2 EXPAND Safety and Effectiveness Results

Thirty-Day Safety Findings

The 30-Day findings in EXPAND demonstrate safe treatment of patients with MR using the MitraClip™ NTR and XTR System. There was a low incidence of major adverse events and single leaflet device attachment events (**Table 24**).

Table 24: Safety Outcomes from EXPAND

	0-30 Days
Major Adverse Events*	3.9% (40/1036)
All-Cause Death	2.3% (24/1036)
Myocardial Infarction	0.0% (0/1036)
Stroke	0.8% (8/1036)
Non-Elective Cardiovascular Surgery for Device-Related Complications	1.1% (11/1036)
Single Leaflet Device Attachment	1.7% (18/1041)

*Subjects are only counted once for each type of event. Subjects who terminated from the study prior to the lower limit of the 30-day visit window (Day 16) without any site reported adverse events are excluded from the analysis.

Thirty-Day Effectiveness Findings

Treatment with the MitraClip™ NTR and XTR System in EXPAND was effective (**Table 25.** There was high acute procedural success rate achieved in 95.8% (985/1028) of subjects with a core lab adjudication of MR ≤2+ at discharge or 30-days without death or reintervention. There was significant MR reduction 88.8% of subjects having an MR ≤1+ at 30-days compared to 8.9% with MR ≤1+ at baseline. Subjects also experienced 30-day symptomatic improvements with 80.1% of subjects in NYHA Class I/II compared to 21.4% at baseline and an increase in KCCQ-OS score of 19.3 ± 25.0 points from baseline to 30 days.

Table 25: Effectiveness Outcomes from EXPAND

	30 Days
MR ≤ 2+	97.8% (845/864)
MR ≤ 1+	88.8% (767/864)
NYHA Class I/II	80.1% (691/863)
Change in KCCQ-OS Score from Baseline	19.3 ± 25.0 (853)

29.2.5 EXPAND G4 Clinical Results

The EXPAND G4 Study is a prospective, multi-center, single-arm, international, post-market, observational study designed to collect real-world data on the use of the next generation MitraClip™ G4 System. The study is designed in two phases:

- Phase I: Clinical and echocardiographic outcomes data was collected from approximately 100 subjects enrolled at approximately 20 sites in the United States with the goal of evaluating the early safety and procedural outcomes associated with the MitraClip™ G4. Phase I was completed August 2020.
- Phase II: Up to 1,000 post-market, consented patients will be treated with a MitraClip G4 device according to local guidelines and IFU from the Europe, the United States, Canada and Japan will be included in the analysis for The MitraClip™ EXPAND G4 Study.

Data collected here reflect the 30-day cutoff of May 9, 2022. Additional results have been published [10].

29.2.5.1 EXPAND G4 Subject Population

From January 21, 2020 to February 4, 2022, a total of 1164 subjects at 60 clinical sites were enrolled into the MitraClip™ EXPAND G4 Study: 101 from Phase I and 1063 from Phase II.

29.2.5.2 EXPAND G4 Safety and Effectiveness Results

Thirty-Day Safety Findings

The 30-Day findings in EXPAND G4 demonstrate safe treatment of patients with MR using the MitraClip™ G4 System. There was a low incidence of major adverse events and single leaflet device attachment events (**Table 26**).

Table 26: Safety Outcomes from EXPAND G4

	0-30 Days
Major Adverse Events*	2.7% (31/1158)
All-Cause Death	1.3% (15/1158)
Myocardial Infarction	0.2% (2/1158)
Stroke	0.5% (6/1158)
Mitral Valve Surgical Reintervention	0.9% (10/1158)
Single Leaflet Device Attachment	1.0% (12/1158)

*Subjects are only counted once for each type of event. Subjects who terminated from the study prior to the lower limit of the 30-day visit window (Day 16) without any site reported adverse events are excluded from the analysis.

Thirty-Day Effectiveness Findings

Treatment with the MitraClip™ G4 System in EXPAND G4 was effective (**Table 27**). There was a high acute procedural success rate achieved in 96.2% (1099/1143) of subjects with a core lab adjudication of MR severity ≤2+ at discharge or 30-days without death or reintervention.

There was significant MR reduction to ≤1+ at 30-days compared to 5.8% of subjects with MR ≤1+ at baseline. Subjects also experienced 30-day symptomatic improvements with a greater proportion in NYHA Class I/II compared to 30.8% at baseline and an increase in KCCQ-OS score of 18.5 ± 24.7 points from baseline to 30-days.

Table 27: Effectiveness Outcomes from EXPAND G4

	30 Days
MR ≤ 2+	97.9% (808/825)
MR ≤ 1+	90.8% (749/825)
NYHA Class I/II	83.3% (862/1035)
Change in KCCQ-OS Score from Baseline	18.5 ± 24.7 (982)

30.0 REFERENCES

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31.0 MITRACLIP™ G5 PROCEDURE ACRONYMS AND DEFINITIONS OF TERMS

31.1 Glossary of Acronyms

Guide:	MitraClip™ G5 Steerable Guide Catheter	RA:	Right Atrium
CDS:	MitraClip™ G5 Delivery System	LA:	Left Atrium
Sleeve:	Steerable Sleeve	LV:	Left Ventricle
DC:	Delivery Catheter	RO:	Radiopaque
Clip:	MitraClip™ G5 Implant	MR:	Mitral Regurgitation

31.2 Definition of Terms

Defined Terms are in italics throughout document.

TERM	DEFINITION AND RELATED TECHNIQUE
Lock the Implant	Rotate the Lock Knob counterclockwise to the tactile locked position. **CAUTION: Confirm the Lock Line Release Latch is NOT lifted before rotating the Lock Knob. Premature Lifting of the Lock Line Release Latch may result in the inability to *Unlock the Implant*.**
Unlock the Implant	Rotate the Lock Knob clockwise to the tactile unlocked position. **CAUTION: Confirm the Lock Line Release Latch is NOT lifted before rotating the Lock Knob. Premature Lifting of the Lock Line Release Latch may result in the inability to *Unlock the Implant*.**
Secure the DC Fastener	Rotate the DC Fastener in the direction of the arrow and “Secure” label until fully secured. Ensure the DC Handle does not rotate while securing DC Fastener. **WARNING: Rotation of the DC Handle or failure to fully secure DC Fastener may result in unwanted Implant Arm orientation changes during grasping and cardiac injury.**
Release the DC Fastener	Rotate the DC Fastener in the opposite direction of the arrow and “Secure” label until released. Ensure the DC Handle does not rotate while releasing DC Fastener. **WARNING: Rotation of the DC Handle may result in unwanted Implant Arm orientation changes during grasping and cardiac injury.**

Open the Implant Arms	1. Confirm the Implant is unlocked. 2. Turn the Arm Positioner at least 1/2 turn in the “Close” (clockwise) direction. 3. Turn the Arm Positioner in the “Open” (counterclockwise) direction until the desired *Implant Arm Angle* is achieved. **NOTE 1:** Ensure the DC handle is held securely when opening the Implant. **NOTE 2:** Implant Arms will begin opening within one rotation of the Arm Positioner. Do not rotate more than one rotation past neutral in the open direction if the Implant Arms do not appear to open. **NOTE 3:** If the Implant Arms fail to open visibly, use the following techniques in the order provided, as needed. Return the Lock Knob to the original unlock position after the Implant is opened to the desired Arm Angle. A. Stop and return *Arm Positioner to Neutral*. Turn Lock Knob to the next Unlock position. B. Incrementally iterate the amount of Arm Positioner rotation in the “Close” direction followed by rotation in the “Open” direction. Iterate until Implant opens. If Implant does not open and it is no longer possible to rotate the Arm Positioner in the “Close” direction, proceed to next step. C. Repeat steps A and B (Lock Knob rotation followed by Arm Positioner iterations) until Implant opens. If Implant does not open, proceed to next step in the sequence. D. *Lock the Implant*. *Lower the Grippers*, *Unlock the Implant*, and repeat **NOTE 3**, Step C. *Raise the Grippers* after Implant opens. If Implant does not open, proceed to step E. E. If the Implant is free of leaflets, then: *Lock the Implant*, release the DC Fastener, release the Sleeve curves, and *Unlock the Implant*. Repeat **NOTE 3**, Step C. **WARNING: Do release the DC Fastener before releasing Sleeve curves, otherwise it may result in device damage and / or device or component embolization.** If the Implant does not open after performing all steps in NOTE 3, DO NOT use the device.
Arm Positioner to Neutral	Turn the Arm Positioner in the “Close” or “Open” direction until no resistance to turning is noted.

Invert the Implant Arms	1. Confirm the Implant is unlocked. 2. Turn the Arm Positioner at least 1/2 turn in the "Close" direction. 3. Turn the Arm Positioner in the "Open" direction until an *Implant Arm Angle* of 180° is observed under fluoroscopic guidance. Note the orientation of the grey line on the Arm Positioner. 4. Continue turning the Arm Positioner in the "Open" direction until the Implant Arms invert, no more than 1 full turn from 180°. DO NOT over-invert the Implant Arms. DO NOT turn Arm Positioner more than 1 full turn past an *Implant Arm Angle* of 180° or past when resistance is first noted. **WARNING: Turning the Arm Positioner in the "Open" direction more than 1 full turn past an *Implant Arm Angle* of 180° or turning past when resistance is first noted may result in device damage which could cause the Implant to become non-functional and lead to embolization, and / or conversion to surgical intervention.**
Raise the Gripper(s)	(a) Simultaneously- Confirm Gripper Levers are latched. Slowly retract the Gripper Levers until a hard stop is reached (under fluoroscopic and echocardiographic observation). or (b) Independently- Unlatch the Gripper Levers and slowly retract the desired Gripper Levers until a hard stop is reached (under fluoroscopic and echocardiographic observation). Re-latch the Gripper Levers.
Lower the Gripper(s)	(a) Simultaneously- Confirm the Gripper Levers are latched and fully advance the Gripper Levers. or (b) Independently- Unlatch the Gripper Levers and fully advance the desired Gripper Lever(s). Re-latch the Gripper Levers.
Identify Gripper Orientation	1. Unlatch the Gripper Levers. 2. Advance and retract the Gripper Lever with the tactile marker under imaging (echocardiography) to identify Gripper Lever to the corresponding leaflet. 3. Repeat step 2 with the Gripper Lever without the tactile marker. 4. Once Grippers are identified, Raise the Gripper(s) and relatch.
Implant Arm Angle	• Angle between the inner edges of both Implant Arms. • All Implant Arm Angles are measured using fluoroscopy with optimal view allowing clear observation of the tip of the Implant and both arms in the same plane, so they appear as a "V" (see Figure 1). All Implant Arm Angles are approximate.
Grasping Arm Angle	An *Implant Arm Angle* of approximately 120 degrees.

Close the Implant Arms	Turn the Arm Positioner in the “Close” direction until the desired *Implant Arm Angle* is achieved per the instructed IFU step. To Fully Close the Implant: • Under direct visualization, the Implant is fully closed when the inner edges of the Implant Arms contact the DC. • Under fluoroscopic observation, the Implant is fully closed when the inner edges of the Implant Arms are parallel. **NOTE**: Ensure the DC handle is held securely when closing the Implant. **NOTE**: If Implant Arms fail to close visibly (as observed under fluoroscopic guidance), use advanced closing techniques as needed.
Test the Lock	Verification step to confirm that the pre-deployment *Implant Arm Angle* will reflect the *Implant Arm Angle* post-deployment. With the Lock Knob pointing distally in the Locked Position, and the Arm Positioner to Neutral (note the orientation of the grey line on the Arm Positioner), turn the Arm Positioner 1 turn in the “Open” direction (confirm grey line has returned to the original orientation). The Implant Arms may open slightly (~5°) and then remain in a stable position. **NOTE**: Ensure the DC handle is held securely when turning the Arm Positioner. **NOTE**: If continued opening of the Implant Arms is noted, reconfirm that the Lock Knob is pointing distally in the Locked Position. *Close the Implant Arms* and *Test the Lock*. **WARNING: DO NOT turn the Arm Positioner more than 1 turn in the “Open” direction from neutral. Failure to stop turning the Arm Positioner at 1 turn in the “Open” direction past neutral may result in Implant opening or device damage which could cause the Implant to become non-functional and lead to embolization and / or conversion to surgical intervention.**
Lift the Lock Line Release Latch	With the Lock Knob pointing distally in the Locked Position, and the Arm Positioner on the closed side of Neutral, Lift the Lock Line Release Latch to the hard stop. **WARNING: Once the latch is lifted, be careful to not inadvertently lower the latch prior to Lock Line removal. DO NOT Lift the Lock Line Release Latch until deployment. Premature Lifting of the Lock Line Release Latch may result in the inability to *Unlock the Implant*.**

Remove the Lock Knob	Retract the Lock Knob Release Slider, carefully detach the Lock Knob from the DC Handle, and release the Lock Knob Release Slider. Carefully pull the Lock Knob until the entire Lock Line is removed from the DC Handle. **WARNING: If resistance is felt during Lock Knob removal, stop and confirm that the Lock Line Release Latch is fully lifted to the hard stop before continuing with Lock Knob removal.**
Left Atrial Pressure Capability	If using left atrial pressure capability, connect a fluid filled pressure monitoring system (not supplied) to the luer port of the Guide valve housing. Calibrate the pressure monitoring system and obtain measurements via the manufacturer's instructions for use. **NOTE:** Ensure no bubbles are present after inserting the CDS into the Guide. When measuring left atrial pressure, ensure the Implant is not obstructing the Guide Lumen, the Guide RO Tip Ring is between the RO Alignment Markers of the Sleeve, and the Sleeve is not deflected more than 90 degrees as this may impair Guide lumen patency.
Secure the DS Attach	Insert the Delivery System (DS) Attach into the Stabilizer Rail. Confirm the DS Attach is secure and stable within the Stabilizer Rail.
Release the DS Attach	Pinch, securely hold, and lift the DS Attach to release it from the Stabilizer Rail.

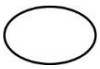
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GRAPHICAL SYMBOLS FOR MEDICAL DEVICE LABELING

Source reference for recognized symbols retained at the manufacturer in the Medical Device File.

	Batch code		Do not resterilize
	Catalogue number		Do not re-use
	Use-by date		Non-pyrogenic
	Sterilized using ethylene oxide		Do not use if package is damaged and consult instructions for use
	Single sterile barrier system		Manufacturer
	Consult instructions for use or consult electronic instructions for use		Date of manufacture
	Refer to instruction manual/booklet (Symbol Color: Blue)		Packaging unit
	MR Conditional		Keep away from sunlight
	Unique device identifier		Keep dry
	Medical device		CAUTION: Federal law restricts this device to sale by or on the order of a physician.